LESSON BOOKLET • YEARS 7 AND 8

Forces and Motion

Types of force • Newton's laws • energy • levers • projectiles

Victorian Curriculum F–10 Version 2.0 | VC2S8U13 | VC2S8U14 | VC2S8U15 | VC2S8I01–I08

Name: _____ **Class:** _____

Teacher: _____ **Year level:** _____

How to use this booklet

There is one section for each lesson, and every lesson works the same way:

- **Part A — Reading** a short passage and comprehension questions to start the lesson
- **Part B — Theory** the ideas, with worked examples to follow step by step
- **Part C — Practice** questions like the examples, so you can build the skill
- **Part D — Above level** harder, higher-order questions — Year 8 and extension

Experiment lessons follow the same four parts: a reading about the science, planning the experiment, results and analysis, then above-level questions. Everything leads to the trebuchet investigation at the end of the unit.

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1 What is a Force?

Pushes and pulls • Measuring force • Mass and weight • Contact and non-contact forces

Victorian Curriculum 2.0: **VC2S8U14 • VC2S8I04**

Part A • Reading and comprehension

Reading — The Forces You Cannot See

Right now, while you sit perfectly still, you are being pushed and pulled from several directions at once. Gravity is pulling every part of your body downwards towards the centre of the Earth. At the same time, the chair beneath you is pushing upwards on you with exactly the same strength. Because these two forces are equal and opposite, you stay put. If the chair suddenly vanished, the pushing force would disappear, gravity would win, and you would fall.

People often think that astronauts on the International Space Station float because there is no gravity in space. This is not true. At the height the Space Station orbits, Earth's gravity is still about 90% as strong as it is on the ground. Astronauts float because they — and the whole Space Station — are falling around the Earth at enormous speed, and they fall together. Everything inside falls at the same rate, so nothing presses against anything else. The astronauts do not feel the push of a floor, so they seem weightless.

Forces are invisible, so scientists need a way to record them. The solution is simple but powerful: draw each force as an arrow. The direction of the arrow shows which way the force acts, and the length of the arrow shows how strong it is. A picture full of invisible pushes and pulls becomes something you can measure, compare and calculate with.

C1. According to the passage, name the two forces acting on you while you sit still, and state the direction of each.

C2. The passage says that if the chair vanished, "gravity would win". Explain what this means, using the idea of two forces that are no longer equal.

C3. Explain why astronauts on the Space Station appear to float, even though gravity is still acting on them.

C4. Does an astronaut on the Space Station still have weight? Use a number from the passage to support your answer.

C5. The passage says drawing forces as arrows is “simple but powerful”. What two pieces of information does a force arrow give you?

C6. Which three-word phrase in the first paragraph describes two forces that are the same size but act in opposite directions?

C7. Forces are invisible. Describe one way you could tell that a force is acting on an object, even though you cannot see the force itself.

C8. Imagine the Space Station suddenly stopped moving around the Earth. Using the passage, predict what would happen to the station and the astronauts inside, and explain why.

Part B • Theory and examples

1.1 Every Force is a Push or a Pull

This is the whole idea, and it really is that simple. Whenever one object affects another — by touching it, or from a distance — it does so by **pushing** it or **pulling** it. There is no third option.

Think about it

A magnet *pulls* a paperclip. A hand *pushes* a door. The Earth *pulls* you downwards. A rope *pulls* a sledge. The ground *pushes* up on your feet.

1.2 What Forces Can Do

A force can only change an object in a small number of ways. Learn these five and you can describe almost any situation in this unit.

#	A force can…	Example
1	Make a still object start moving	A kicked football leaves the ground
2	Make a moving object stop	Brakes bring a bicycle to a halt
3	Make an object speed up or slow down	Pedalling harder makes you accelerate
4	Change the direction of motion	A tennis racquet sends the ball back
5	Change the shape of an object	Squashing a drink can

1.3 Measuring Force: The Newton

Force is measured in **newtons**, written **N**. The unit is named after Sir Isaac Newton, whose laws of motion you will meet in Lesson 4.

Getting a feel for a newton

1 N is roughly the force you feel holding a small apple (about 100 g) against gravity.
10 N is about the weight of a 1 kg bag of sugar.
500 N is roughly the weight of an average adult.

We measure force with a **spring scale** (also called a *newton meter*). Inside is a spring: the harder you pull, the further it stretches, and a pointer moves down a scale marked in newtons.

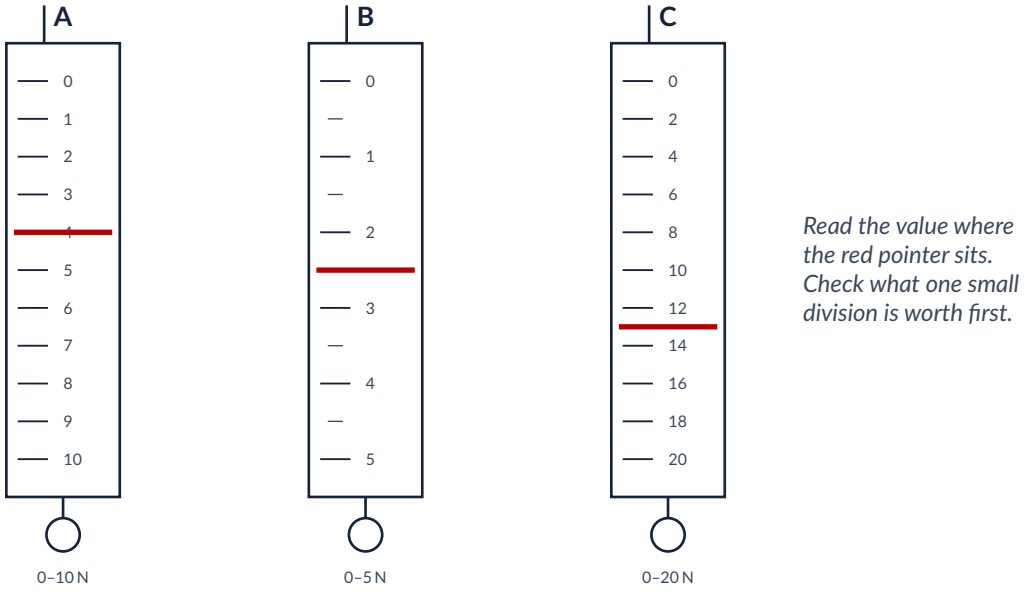


Figure 1.1 — Reading a spring scale. Always read at eye level, and check the size of each small division first.

How to read a spring scale correctly

1. Check the **largest** number on the scale — this is the maximum it can measure.
2. Work out what **one small division** is worth. If there are 10 small gaps between 0 and 5 N, each is 0.5 N.
3. Hold the scale still and read at **eye level** to avoid parallax error.
4. Read the value where the **pointer** sits, and always write the **unit (N)**.

Worked Example 1.1 — Reading a spring scale

A spring scale is marked from 0 to 10 N, with **20** small divisions between 0 and 10. The pointer sits on the **7th** small division. What is the reading?

Step 1 — one division: $10 \div 20 = 0.5$ N per division
Step 2 — the reading: $7 \times 0.5 = \mathbf{3.5}$ N

1.4 Mass and Weight Are Not the Same Thing

In everyday speech we use these words as if they mean the same thing. In science they are quite different, and mixing them up causes a lot of lost marks.

	Mass	Weight
What it is	The amount of matter in an object	The *force* of gravity pulling on that mass
Unit	kilograms (kg)	newtons (N)
Measured with	A balance	A spring scale
Changes?	Same everywhere in the universe	Changes with the strength of gravity

The link between them

$$\text{Weight} = \text{mass} \times \text{gravitational field strength} \quad \Rightarrow \quad \boxed{W = mg}$$

On Earth, $g = 9.8$ N/kg. On the Moon, $g = 1.6$ N/kg.

Worked Example 1.2 — Mass and weight

A student has a mass of 50 kg. Find their weight on Earth, and on the Moon.

On Earth: $W = mg = 50 \times 9.8 =$ **490** N
On the Moon: $W = mg = 50 \times 1.6 =$ **80** N

Their *mass* is 50 kg in both places — only the *weight* changes.

1.5 Contact and Non-contact Forces

Forces fall into two groups. **Contact forces** only act when two objects touch. **Non-contact forces** act across a gap — the objects do not need to touch at all.

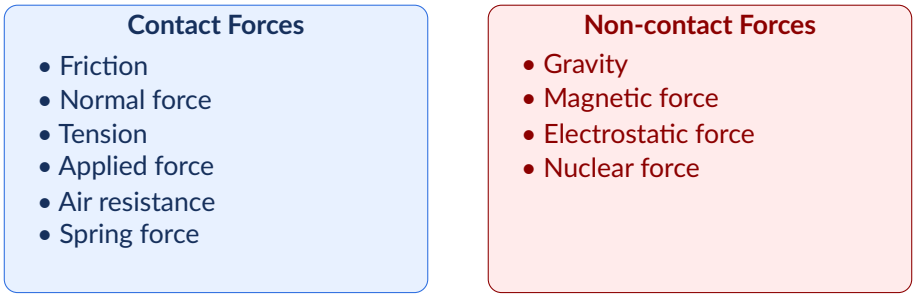


Figure 1.2 — Contact and non-contact forces

Worked Example 1.3 — Classifying forces

A fridge magnet holds a shopping list on the fridge door without sliding down. Name two forces acting on the magnet and classify each one.

Weight pulls the magnet down — gravity acts across a gap, so it is a **non-contact** force.
Friction between the magnet and the door stops it sliding — the surfaces must touch, so it is a **contact** force.

(The magnetic pull towards the door is also non-contact — it is what presses the magnet on hard enough for friction to hold it.)

Notes

Part C • Practice

Work through every task. Show your working for calculations, and always include the unit.

Task 1.1

For each everyday action, decide whether the force is a push or a pull. Write **PUSH** or **PULL** in the box.

Action	Push or pull?	Action	Push or pull?
Kicking a football	*PUSH*	Opening a drawer	*PULL*
A crane lifting a girder		Squeezing a stress ball	
A dog straining on a lead		Wind on a sail	

Task 1.2

For each event below, write the **number** of the effect the force is having. Some events have more than one.

1	2	3	4	5
starts it moving	stops it	speeds it up or slows it down	changes its direction	changes its shape

Event	Effect (1–5)	Event	Effect (1–5)
A goalkeeper catches a ball	*2*	Stretching a rubber band	*5*
A car turns a corner at steady speed		A rocket lifts off	

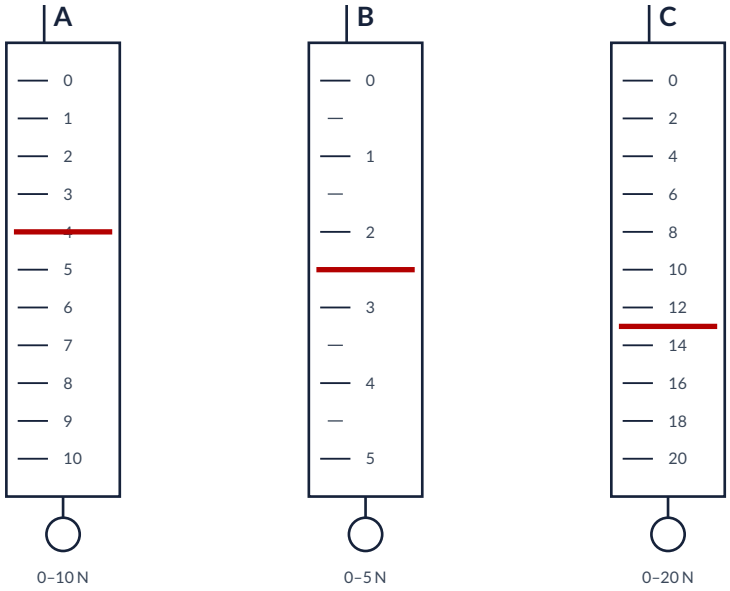
Task 1.3

Think of three forces you used this morning. For each one, describe the action, write **PUSH** or **PULL**, and give the number of the effect it had: **1** starts it moving, **2** stops it, **3** speeds it up or slows it down, **4** changes its direction, **5** changes its shape.

Action	Push or pull?	Effect (1–5)
Opening the fridge door	*PULL*	*1*

Task 1.4

Write the reading shown on each of the three spring scales below. Check what one small division is worth on each scale first.



Read the value where the red pointer sits. Check what one small division is worth first.

Scale A	Scale B	Scale C

Task 1.5

Complete the table. Use $g = 9.8$ N/kg on Earth and $g = 1.6$ N/kg on the Moon.

Object	Mass (kg)	Weight on Earth (N)	Weight on Moon (N)
Bag of sugar	1	$1 \times 9.8 = 9.8$	$1 \times 1.6 = 1.6$
Bicycle	12		
Small car	900		

Task 1.6

A bag of flour has a mass of 2 kg. Find its weight on Earth. Use $g = 9.8$ N/kg.

Task 1.7

Which has the greater weight on Earth: a 5 kg dog, or a box that weighs 50 N? Show your working.

Task 1.8

Complete the table. For each force, write whether it is a contact or non-contact force, and give a real-world example.

Force	Contact / Non-contact	Real-world example
Gravity	*Non-contact*	*An apple falling from a tree*
Friction		
Magnetic force		
Normal force		
Air resistance		
Tension		

Task 1.9

A spring scale is marked from 0 to 20 N, with **40** small divisions. The pointer sits on the **13th** small division. What is the reading? Show both steps.

Task 1.10

A cat has a mass of 4 kg. Find its weight on Earth.

Task 1.11

The same cat travels to the Moon. Find its weight there. Has its mass changed?

Task 1.12

A balloon rubbed on your hair sticks to a wall. Name the force holding it there, and say whether it is a contact or non-contact force.

Part D • Above level

These questions go beyond the Year 7 standard. They ask you to combine ideas, use a formula, or judge someone else's reasoning.

Extension 1.13
An astronaut has a mass of 75 kg.
(a) Find her weight on Earth.
(b) Find her weight on the Moon.
(c) Explain why she can jump much higher on the Moon, even though her mass has not changed.

Extension 1.14
A suitcase has a weight of 196 N on Earth.
(a) Find its mass.
(b) Find its weight on the Moon.

Extension 1.15
A bathroom scale really measures *force*, but it is labelled in kilograms. A 60 kg student reads 60 kg on it on Earth. Predict what the same scale would read on the Moon, and explain why the scale is not really measuring mass.

Extension 1.16
On Mars, $g = 3.7$ N/kg. A 10 kg backpack is carried on Earth ($g = 9.8$ N/kg), on the Moon ($g = 1.6$ N/kg) and on Mars.
(a) Calculate the backpack's weight in each place.
(b) Rank the three places from easiest to hardest to carry the backpack, and explain your ranking.

Extension 1.17
A rock hangs from a spring scale on Earth, and the scale reads 4.9 N. Use $g = 9.8$ N/kg on Earth and $g = 1.6$ N/kg on the Moon.
(a) Find the mass of the rock.
(b) Find what the same spring scale would read if the rock were hung from it on the Moon.

Extension 1.18
A *balance* compares an object with known masses. A *spring scale* measures the pull of gravity on the object. Explain why a balance gives the same result on the Moon as on Earth, but a spring scale does not.

Extension 1.19
Magnets can push or pull without touching. Describe a quick test you could do with a magnet, a paperclip and a sheet of paper to show that magnetic force acts *through* the paper. Say what you would do and what you would expect to see.

2 Force Diagrams and Net Force

Naming forces • Force arrows • Free body diagrams • Balanced and unbalanced forces

Victorian Curriculum 2.0: **VC2S8U14 • VC2S8I04**

Part A • Reading and comprehension

Reading — Winning a Tug-of-War

In a tug-of-war, two teams pull on the same rope in opposite directions. While the rope stays still, the two pulls are exactly the same size. They are *balanced*, so the net force on the rope is zero — the rope does not move, even though both teams are pulling as hard as they can.

The moment one team pulls even a little harder, the forces become *unbalanced*. There is now a net force towards the stronger team, and the rope starts to move their way. The bigger the difference between the two pulls, the faster the rope speeds up.

So how does a team pull harder? Strength helps, but the real secret is under their feet. To pull on the rope, each person pushes backwards against the ground with their shoes. Friction between the shoes and the ground stops them from sliding forwards. A team with good grip — rough soles, a dry surface, bodies low and leaning back — can pull harder before their feet slip. A team on a slippery surface will be dragged across, no matter how strong they are.

Coaches explain this with force arrows. They draw the rope, the two pulls and the friction on the feet, each as an arrow whose length shows its size. One look at the arrows shows which team is going to win.

C1. While the rope is not moving, what can you say about the two teams' pulls? Use the word *balanced*.

C2. What happens to the rope when one team pulls harder? Explain why, using the words *net force*.

C3. According to the passage, why does a team on a slippery surface lose, however strong they are?

C4. List two things from the passage that give a team better grip.

C5. How can a coach tell which team will win just by looking at the force arrows?

C6. Explain why each person needs friction under their shoes in order to pull on the rope.

C7. A team plays in socks on a polished wooden floor. Use the passage to predict what will happen to them, and explain why.

C8. In your own words, explain the difference between balanced and unbalanced forces. Give an example of each from the passage.

Part B • Theory and examples

2.1 Naming the Forces

Before you can draw a force diagram, you need the right names. These are the forces you will use again and again in this unit.

Force	What causes it	Which way it acts
Weight (W)	Gravity pulling on mass	Always straight down
Normal force (N)	A surface pushing back on an object	Perpendicular to the surface
Friction (f)	Two surfaces rubbing together	Opposite to the motion
Air resistance	Air pushing against a moving object	Opposite to the motion
Tension (T)	A rope, string or cable pulling	Along the rope, away from the object
Applied force	A person or machine pushing/pulling	Whichever way it is applied

2.2 Drawing Force Arrows — The Three Rules

The three rules for every force arrow

1. **Start the arrow on the object.** The tail sits on the object the force acts on.
2. **Point the arrowhead the way the force acts.** Down for weight, up for the normal force, and so on.
3. **Make the length match the size.** A 20 N force is drawn twice as long as a 10 N force. *Always.*

Rule 3 is the one students most often forget — and it matters, because the picture should tell you the answer at a glance. If two arrows are the same length, the forces are balanced. If one is longer, that is the direction the object accelerates.

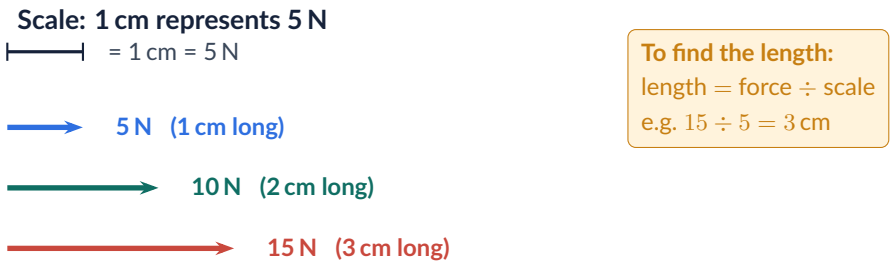


Figure 2.1 — Using a scale for force arrows. Here 1 cm represents 5 N, so a 15 N force is drawn 3 cm long.

Worked Example 2.1 — Drawing arrows to scale

Using a scale of 1 cm = 5 N, how long should the arrow be for a 25 N push?

Length = force ÷ scale = $25 \div 5 =$ **5** cm

2.3 Free Body Diagrams

A **free body diagram (FBD)** shows *all* the forces acting on *one* object, each drawn as a labelled arrow using the three rules. The object is drawn as a simple box or dot — nothing else from the picture is included.

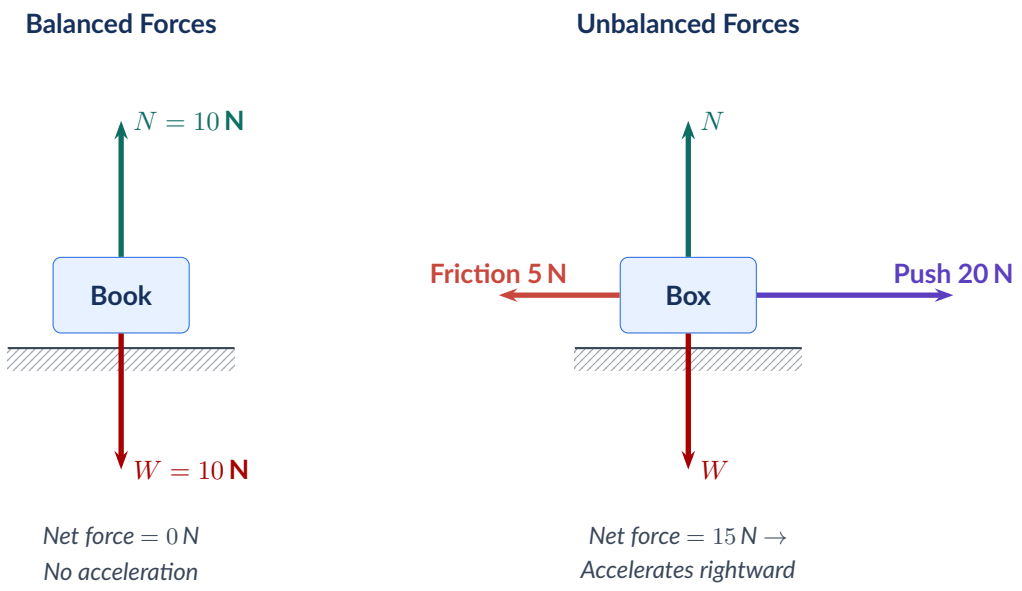


Figure 2.2 — Balanced forces (left) and unbalanced forces (right)

Worked Example 2.2 — Drawing a free body diagram

A book with a weight of 10 N rests on a table. Draw its free body diagram.

Step 1 — find every force. Gravity pulls the book down (*weight*). The table pushes up on it (*normal force*).
Step 2 — size each force. The book is not moving up or down, so the table pushes up exactly as hard as gravity pulls down: both are 10 N.
Step 3 — draw. A box for the book, a 10 N arrow down labelled W, and an arrow of the *same length* up labelled N — as in the left of Figure 2.2.

2.4 Balanced and Unbalanced: Net Force

When you add up all the forces on an object, the single force left over is called the **net force** (or resultant force).

Working out the net force

Same direction → **add** the forces.
Opposite directions → **subtract** the smaller from the larger. The net force acts in the direction of the larger force.

If the net force is **zero**, the forces are **balanced** and the motion does not change.
If the net force is **not zero**, the forces are **unbalanced** and the object speeds up, slows down or changes direction.

Worked Example 2.3 — Finding the net force

A sledge is pulled forwards with 45 N. Friction pushes back with 18 N. Find the net force.

Opposite directions, so subtract: $45 - 18 = \mathbf{27}$ N forwards

The net force is not zero, so the forces are unbalanced and the sledge **speeds up**.

Part C • Practice

Work through every task. Show your working for calculations, and always include the unit.

Task 2.1

Using a scale of **1 cm = 2 N**, how long should the arrow be for each force?

(a) 6 N

(b) 10 N

(c) 3 N

Task 2.2

Using a scale of 1 cm = 10 N, how long should the arrow be for a 45 N pull?

Task 2.3

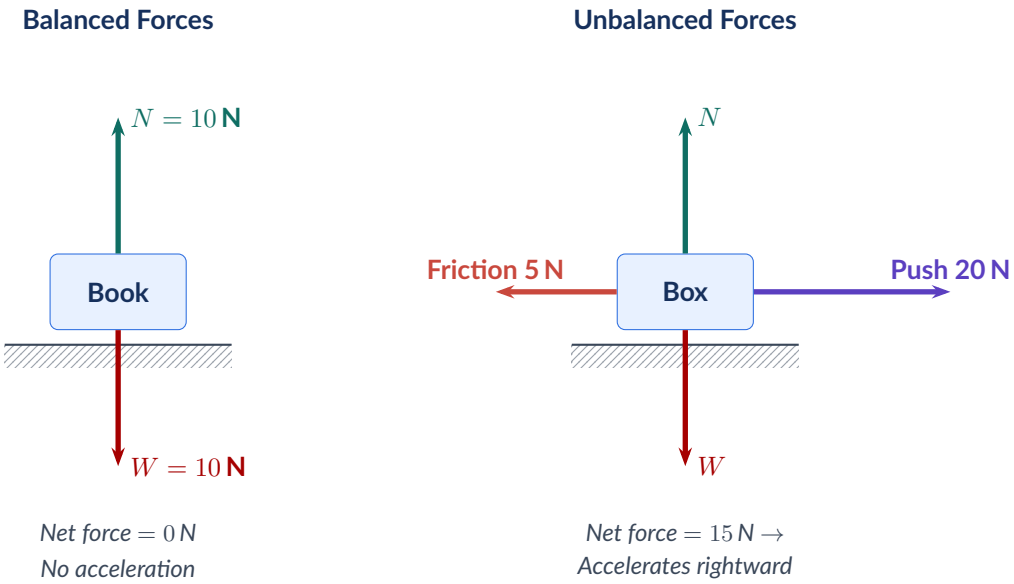
An arrow is 3.5 cm long. The scale is 1 cm = 20 N. What force does it show?

Task 2.4

A box is pulled right with 30 N and friction resists with 10 N. Using a scale of **1 cm = 10 N**, draw both arrows on the box below. Label each with its name and size.

Task 2.5a

The diagrams below show the forces on a book resting on a table (left) and on a box being pushed along the floor (right).



Why is the book not speeding up or slowing down? Use the words *net force* in your answer.

Task 2.5b

The box is pushed right with 20 N, and friction pushes left with 5 N. Calculate the net force on the box and state which way it speeds up.

Task 2.5c

Draw a free body diagram for a skydiver falling at a steady speed (terminal velocity). Label each force with its name and show their comparative sizes.

Draw your force diagram here.

Task 2.6

A lamp with a weight of 15 N hangs still from the ceiling on a cord. Draw its free body diagram, labelling each force with its name and size.

Draw your force diagram here.

Net force. For each question, state the size *and* direction of the net force, and whether the forces are balanced or unbalanced.

Task 2.7

Two people push a car in the same direction with 250 N and 310 N. Find the net force.

Task 2.8

A trolley is pushed right with 40 N while friction pushes left with 12 N. Find the net force.

Task 2.9

A book rests on a table. Its weight is 8 N. What is the normal force from the table? Explain.

Task 2.10

In a tug-of-war, one team pulls left with 900 N and the other pulls right with 900 N. Describe the motion of the rope.

Task 2.11

A rocket engine produces 50 000 N of upward thrust. The rocket's weight is 32 000 N.

(a) Find the net force.

(b) State which way the rocket speeds up.

(c) Explain what would happen if the thrust dropped to 32 000 N.

Task 2.12
A box has three forces on it: 15 N right, 10 N right and 30 N left. Find the size and direction of the net force.

Task 2.13
A skydiver weighs 600 N. Air resistance on her is 600 N. Are the forces balanced or unbalanced? Describe her motion.

Task 2.14
A swimmer moves forward at a steady speed. Her arms provide 120 N of forward force.

(a) What is the size of the water resistance acting on her?

(b) Explain how you know.

Task 2.15
A parachutist has a weight of 700 N. Just after the parachute opens, air resistance is 950 N.

(a) Find the net force and its direction.

(b) Describe what happens to the parachutist's speed.

(c) Air resistance then falls to 700 N — describe the motion now.

Task 2.16
In a tug-of-war, Team A pulls the rope left with 550 N and Team B pulls it right with 620 N.

(a) Find the size and direction of the net force on the rope.

(b) Which team is winning? Explain using the words *unbalanced* and *net force*.

(c) Draw both pulls on the rope below as force arrows, using a scale of 1 cm = 100 N. Label each arrow with its size.



Part D • Above level

These questions go beyond the Year 7 standard. They ask you to combine ideas, use a formula, or judge someone else's reasoning.

Extension 2.17
Sam says: *"A hockey puck sliding across ice at a steady speed must have an unbalanced force pushing it forwards."* Is Sam right? Use a free body diagram and the idea of net force to explain.

Extension 2.18
A skydiver jumps from a plane. Draw a free body diagram for each moment below. Then explain why air resistance grows, and why the skydiver eventually stops speeding up.
(i) just after jumping
(ii) while still speeding up
(iii) at terminal velocity

(i) just after jumping

(ii) speeding up

(iii) terminal velocity

Extension 2.19
A car has a driving force of 4500 N and a total resistive force of 1200 N.
(a) Calculate the net force.
(b) State which way the car accelerates.
(c) Suggest one way the driver could make the forces balanced.

Extension 2.20
A 1200 kg car has a driving force of 4800 N. Air resistance and friction total 1500 N.
(a) Find the net force.
(b) Use $F = ma$ (acceleration = net force ÷ mass) to find the acceleration.
(c) The driving force rises to 6000 N — find the new acceleration.

Extension 2.21
A car drives along a flat road at a steady 60 km/h. Draw a free body diagram for the car showing four forces: weight, the normal force, the driving force and the total resistive force. Show which pairs of forces are balanced.

Draw your force diagram here.

Extension 2.22
A 2 kg ball is falling. Air resistance pushes up on it with 5 N. Use $W = mg$ with $g = 9.8$ N/kg.
(a) Find the weight of the ball.
(b) Find the size and direction of the net force.
(c) Is the ball speeding up, slowing down or moving at a steady speed? Explain.

Extension 2.23
A loaded truck and a small car are both travelling at the same speed. Explain, using the idea of net force, why the truck needs a much bigger braking force to stop in the same distance.

3 Experiment 1: Investigating Friction

Prediction • Fair testing • Repeat trials • Means

Victorian Curriculum 2.0: **VC2S8U14 • VC2S8I01 • VC2S8I02 • VC2S8I03 • VC2S8I04 • VC2S8I05**

Part A • Reading and comprehension

Reading — Grip

Friction is the force that makes walking possible. With every step, your shoe pushes backwards against the ground, and friction pushes you forwards. On ice, where friction is tiny, your foot slips backwards and you go nowhere.

Friction comes from the roughness of surfaces. Even surfaces that look smooth are covered in tiny bumps when you look at them under a microscope. When two surfaces press together, the bumps catch on each other and resist sliding. Rougher surfaces have bigger bumps that catch more, so they produce more friction. Pressing the surfaces together harder also increases friction.

Sometimes engineers want as much friction as possible — in the rubber of a car tyre, the soles of running shoes or the grip tape on a skateboard. Other times they want as little as possible: oil in an engine and wax on skis reduce friction, so parts slide easily and less energy is wasted as heat.

In this experiment you will measure friction with a spring scale. When you pull a block along at a *steady* speed, your pull and the friction are balanced — so the reading on the spring scale is equal to the friction force.

C1. Explain how friction helps you to walk, and why walking on ice is so hard.

C2. Why do rougher surfaces produce more friction? Use the word *bumps* in your answer.

C3. Give one example where engineers want more friction, and one where they want less.

C4. Why must the block move at a *steady* speed when you read the spring scale?

Part B • Planning the experiment

Experiment 1: Investigating Friction

Curriculum: VC2S8U14 • VC2S8I01 • VC2S8I02 • VC2S8I03 • VC2S8I04 • VC2S8I05
Aim: *To investigate how the type of surface affects the friction force on a sliding object.*

3.1 Making it a Fair Test

In a fair test you change **one** thing, measure **one** thing, and keep everything else the same.

Variable	What it means
Independent	The one thing *you* change on purpose
Dependent	The thing you *measure* to see the effect
Controlled	Everything you keep the *same*, so the test is fair

Worked Example 3.1 — Identifying variables

A student rolls a toy car down a ramp at different heights and measures how far it travels across the floor. Identify the variables.

Independent: the height of the ramp (the student changes it).
Dependent: the distance the car travels (the student measures it).
Controlled: the same car, the same floor surface, releasing the car without pushing it.

Prediction — before you begin

Which surface do you think will produce the most friction? Why?

Equipment

Wooden block (or heavy book) Spring scale (newton meter) 3+ surfaces (bench, sandpaper, fabric, carpet) Ruler

Method

1. Place the wooden block on the first surface.
2. Attach the spring scale to the block. Pull slowly and steadily.
3. Read the spring scale while the block moves at a steady speed. Record the friction force in newtons.
4. Repeat 3 times for this surface and record all values.
5. Repeat steps 1–4 for each surface.

Safety

Pull gently and keep the scale in line with the block — a spring scale that is overstretched can snap back. Keep sandpaper away from eyes and skin.

Notes

Part C • Results and analysis

Results

Surface	Trial 1 (N)	Trial 2 (N)	Trial 3 (N)	Mean (N)
e.g. Bench	*1.2*	*1.4*	*1.3*	*1.3*

Calculating a mean

Add the trials, then divide by how many there are. For 1.2 N, 1.4 N and 1.3 N: $(1.2 + 1.4 + 1.3) \div 3 = 3.9 \div 3 =$ **1.3** N

Task 3.1

Identify the independent variable, the dependent variable, and two controlled variables in *your* experiment.

Task 3.2

Which surface produced the greatest friction? Which produced the least? Was your prediction correct?

Task 3.3

Why did you take three trials for each surface? What did you do with the three values?

Task 3.4

Another group measured 2.1 N, 2.3 N and 2.2 N on carpet. Calculate the mean.

Task 3.5

A group measured 1.8 N, 1.9 N and 2.9 N on fabric. Which value looks wrong? Find the mean of the other two.

Task 3.6

Rough surfaces are covered in bigger bumps, which catch on each other and resist sliding. Use this idea to explain *why* the surface that gave the most friction in your experiment did so.

Task 3.7

Draw a column graph of the *mean* friction force for each surface you tested. Put the surfaces on the horizontal axis and **friction force (N)** on the vertical axis. Give your graph a title.

Part D • Above level

These questions go beyond the Year 7 standard. They ask you to evaluate the method, predict, or use a formula.

Extension 3.8
Even on the same surface, your three readings were probably not identical. Suggest one reason why, and one change to the method that would make the readings more consistent.

Extension 3.9
Predict what would happen to the friction force if you stacked a second block on top of the first. Explain your prediction.
Hint: pressing two surfaces together harder makes their bumps catch more.

Extension 3.10
The wooden block has a mass of 0.3 kg. Choose your highest mean friction force. If the block were sliding and you let go of the spring scale, friction would be the only horizontal force on it. Use $F = ma$ (acceleration = force ÷ mass) to calculate how quickly it would slow down. Show all working.

Extension 3.11
Design a fair test to find out whether the *mass* of the block changes the friction force on the bench.
(a) Name the independent and dependent variables.
(b) Name two variables you would keep the same.
(c) Predict what you would find, and give a reason.

4 Newton's First and Third Laws

Inertia • Why moving things keep moving • Action and reaction • Force pairs

Victorian Curriculum 2.0: **VC2S8U14 • VC2S10U17 (above level)**

Part A • Reading and comprehension

Reading — Why Rockets Work in Empty Space

When rockets were first proposed as a way of reaching space, some newspapers scoffed at the idea. In 1920 an editorial in the *New York Times* argued that a rocket could never work beyond the atmosphere, because out in the vacuum there would be nothing for the exhaust gases to push against. It seemed like common sense. A swimmer pushes against water, a runner pushes against the ground — so surely a rocket must push against the air?

The reasoning was wrong, and the mistake came from misunderstanding Newton's Third Law. A rocket does not push against the air behind it. It pushes against *its own exhaust*. The engine hurls a huge mass of hot gas backwards, and by the Third Law the gas pushes the rocket forwards with exactly the same force. Nothing outside the rocket is needed at all. In fact rockets work *better* in space, because there is no air resistance slowing them down.

Out in deep space, something even more surprising happens: a spacecraft can switch its engines off and simply keep going. Far from any planet there is no air and almost nothing pushing or pulling on it, so there is almost no unbalanced force to slow it down. The two *Voyager* space probes were launched in 1977. No engine has pushed them forwards for more than thirty years, yet they are still travelling away from the Sun at more than 55 000 kilometres per hour. Here on Earth, a ball rolled along the ground soon stops — but only because friction and air resistance act on it. Take those away, and a moving object just keeps moving.

The *New York Times* eventually printed a correction — in July 1969, the day after Apollo 11 launched for the Moon, and forty-nine years after the original editorial. It noted, briefly, that further investigation had confirmed the work of Newton in the seventeenth century, and regretted the error.

C1. What mistaken reason did the 1920 editorial give for believing rockets could not work in space?

C2. Explain, using Newton's Third Law, what a rocket actually pushes against.

C3. The passage says rockets work *better* in space. Give the reason, and name the force that is absent.

C4. The *Voyager* probes are still moving fast, even though no engine is pushing them. Explain why they have not slowed down.

C5. According to the passage, why does a ball rolled along the ground on Earth soon stop?

C6. Which phrase in the second paragraph tells you how the push of the gas on the rocket compares with the push of the rocket on the gas?

C7. The editorial's reasoning "seemed like common sense". Using the two examples in the passage, explain why it seemed right to many people.

C8. Imagine a *Voyager* probe drifted into a thin cloud of gas and dust in space. Using the passage, predict what would happen to its speed, and explain why.

Part B • Theory and examples

4.1 Newton's First Law — The Law of Inertia

In Lesson 2 you learned that when the forces on an object are balanced, its motion does not change. Nearly 340 years ago, Sir Isaac Newton made this the first of his three laws of motion.

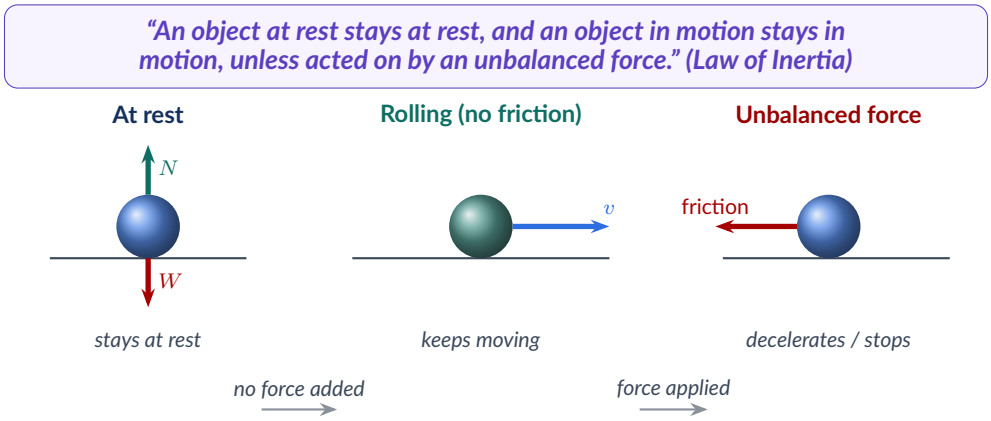


Figure 4.1 — Newton's First Law

First Law

"An object at rest stays at rest, and an object in motion stays in motion at constant velocity, unless acted upon by an unbalanced (net) force."

Inertia is the tendency of an object to resist changes to its motion. Greater mass = greater inertia.

The First Law has two halves, and both say the same thing: *without an unbalanced force, nothing about the motion changes.*

If the object is…	With no unbalanced force, it…	Example
At rest	stays at rest	A book on a shelf never starts sliding by itself
Moving	keeps moving at the same speed, in the same direction	A space probe coasting with its engines off

Think about it — why do things on Earth seem to stop by themselves?

For almost 2000 years, people followed the Greek thinker Aristotle, who taught that a moving object needs a force to *keep* it moving. It seems right: a ball rolled along the ground soon stops. Newton showed that the ball stops because **friction** and **air resistance** are acting on it. They are unbalanced forces, and they slow it down. With no friction and no air resistance, the ball would roll on forever.

Mass and inertia

The more mass an object has, the more inertia it has — so the harder it is to change its motion in *any* way.

	Empty shopping trolley	Full shopping trolley
Mass and inertia	Small	Large
Getting it moving	Easy	Hard
Stopping it	Easy	Hard
Steering it round a corner	Easy	Hard

Worked Example 4.1 — At rest stays at rest

A coin sits on a playing card on top of a glass. The card is flicked quickly sideways. Explain why the coin drops into the glass instead of flying off with the card.

Step 1 — before the flick. The coin is at rest. Its weight pulls down and the card pushes up: the forces are balanced.
Step 2 — the flick. The big force acts on the *card*, not the coin. Only a tiny friction force acts on the coin, for a very short time.
Step 3 — inertia. The coin's inertia keeps it at rest where it is. With the card gone, nothing pushes up on the coin, so its weight is now **unbalanced** and it falls straight into the glass.

Worked Example 4.2 — Moving keeps moving

A car travelling at 50 km/h crashes into a wall and stops suddenly. Explain what happens to a passenger who is not wearing a seatbelt, and how a seatbelt helps.

Step 1 — what the crash does. The wall exerts a huge unbalanced force on the *car*, so the car stops.
Step 2 — inertia. No force has acted on the passenger yet, so by the First Law the passenger **keeps moving forwards at 50 km/h** — until they hit the dashboard or windscreen.
Step 3 — the seatbelt. A seatbelt supplies the unbalanced force that slows the passenger down *with* the car, spread across the strong bones of the chest and hips.

4.2 Newton's Third Law — Action and Reaction

Forces always come in pairs. You cannot push on something without it pushing back on you. Press your hand flat against a wall and you can feel the wall pushing back.

Third Law

"For every action force, there is an equal and opposite reaction force."

Action–reaction pairs **always act on different objects** — they do **not** cancel each other.

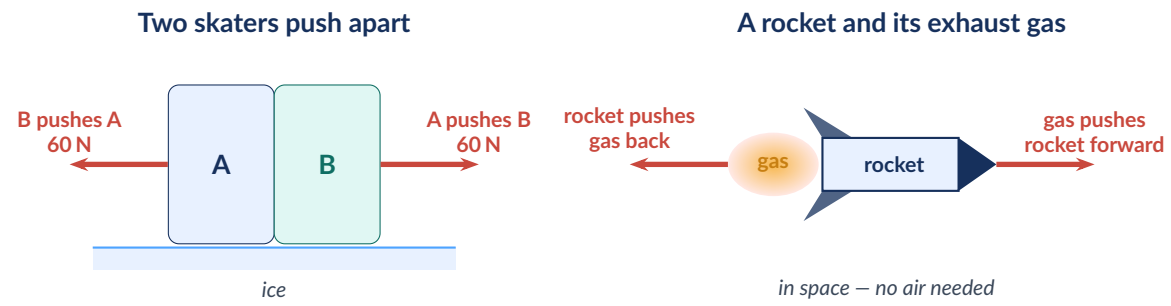


Figure 4.2 — Two Third Law pairs. In each pair the forces are the same size, point in opposite directions and act on different objects.

How to find the reaction force

Write the action as "*A pushes B*", then swap the names: the reaction is "*B pushes A*". The reaction force is always:

1. the **same size** as the action force;
2. in the **opposite direction**;
3. acting on the **other object**;
4. the **same type** of force — a push answers a push, and a gravity pull answers a gravity pull.

Action force	Reaction force
Your foot pushes the ground backwards	The ground pushes your foot forwards
A swimmer's hands push the water backwards	The water pushes the swimmer's hands forwards
A rocket pushes its exhaust gas backwards	The gas pushes the rocket forwards
The Earth pulls you down (your weight)	You pull the Earth up

Why don't the two forces cancel?

Forces only cancel when they act on the *same* object. A Third Law pair always acts on *two different* objects, so the forces can never cancel. When the skaters in Figure 4.2 push apart, each skater has one unbalanced force acting on them — so *both* skaters move, in opposite directions. If one skater has less mass, they move away faster. Lesson 5 shows why.

Worked Example 4.3 — Describing a reaction force

A student on roller skates pushes against a wall with a force of 40 N. Describe the reaction force: name it, and give its size and direction. What happens to the student?

Action: the student pushes on the wall, 40 N, towards the wall.
Reaction: the **wall pushes on the student**, with **40 N**, **away from the wall**.
Motion: this force acts on the student, and the skate wheels give very little friction, so it is **unbalanced**. The student rolls backwards, away from the wall.

Notes

Part C • Practice

Work through every task. Use the words *inertia*, *unbalanced force*, *action* and *reaction* wherever they fit.

Task 4.1

For each situation, decide which law explains it. Write **FIRST** or **THIRD** in the box.

Situation	Law	Situation	Law
Passengers lurch forwards when a bus brakes	*FIRST*	A swimmer pushes water back and moves forwards	*THIRD*
A puck keeps sliding across smooth ice		A balloon flies off when its air rushes out	
A tablecloth is whipped away and the plates stay put		A fire hose pushes back hard on the firefighters holding it	
When a car turns sharply right, you slide to the left of your seat		Pushing off the side of a pool sends you away from the wall	

Task 4.2

Explain why passengers lurch forwards when a bus suddenly brakes. Use the words *inertia* and *unbalanced force* in your answer.

Task 4.3

A hockey puck slides across an almost frictionless ice surface.

(a) Describe how the puck moves while it slides.

(b) What will eventually stop it, and why?

Task 4.4

Which has more inertia: an empty shopping trolley or a full one? Explain.

Task 4.5

A truck and a small car travel at the same speed. Why does the truck need much stronger brakes?

Task 4.6
The same push sends a ball rolling across carpet and across a smooth wooden floor. On which surface does it roll further? Explain.

Task 4.7
When you shake a wet umbrella and then stop it suddenly, the water flies off. Use the idea of inertia to explain why.

Task 4.8
When a car is hit hard from behind, it is suddenly pushed forwards. Use the First Law to explain why a passenger's head is jolted backwards, and how a headrest helps.

Task 4.9
Complete the table. For each action force, describe the reaction force.

Action force	Reaction force
A boat's paddle pushes the water backwards	*The water pushes the paddle (and the boat) forwards*
A hammer pushes down on a nail	
A tennis racquet pushes the ball forwards	
A bird's wings push the air downwards	
The Earth pulls an apple downwards	

Task 4.10
A swimmer pushes backward against the water with a force of 80 N. Describe the reaction force: name it, state its size, and state its direction.

Task 4.11

Two skaters stand still on ice, facing each other. Skater A pushes skater B with a force of 30 N. Using a scale of **1 cm = 10 N**, draw both forces of the Third Law pair. Draw each skater as a box, start each arrow on the skater it acts on, and label each arrow with its size.

Draw your force diagram here.

Task 4.12

When you blow up a balloon and let it go, it zooms around the room.

(a) What does the balloon push on?

(b) Use the Third Law to explain why the balloon moves forwards.

Task 4.13

A skateboarder jumps forwards off the back of a skateboard, and the skateboard shoots off backwards. Explain why, naming the action force and the reaction force.

Task 4.14

A student steps forwards off a small rowing boat onto a jetty, and the boat drifts backwards, away from the jetty.

(a) Name the action force and the reaction force.

(b) Explain why the boat moves, even though nobody pushed it with their hands.

Part D • Above level

These questions go beyond the Year 7 standard. They ask you to combine ideas, use both laws together, or judge someone else's reasoning.

Extension 4.15
Jess says: *"The weight of a book on a table and the normal force from the table are a Third Law pair, because they are equal and opposite."* Is Jess right? Explain, and name the true reaction force to each of the two forces.

Extension 4.16
For each pair of forces, decide whether it is a Third Law pair. Write **YES** or **NO**, and explain how you decided.
(i) The Earth pulls a book down; the book pulls the Earth up.
(ii) A rope pulls a sledge forwards; friction from the snow pulls the sledge backwards.
(iii) A swimmer pushes the water backwards; the water pushes the swimmer forwards.

Extension 4.17
A horse pulls a cart. By the Third Law, the cart pulls back on the horse with an equal force. Liam says: *"So the forces cancel out, and the horse can never get the cart moving."* Explain Liam's mistake. *(Hint: draw a free body diagram for the horse alone. What pushes the horse forwards?)*

Extension 4.18
An astronaut on a spacewalk is floating 10 m from the Space Station, not moving. Her safety line has come loose, but she is holding a heavy toolbox.
(a) Explain how she can use the Third Law to start moving back towards the station.
(b) Once she is moving, does she need to keep doing anything to keep moving? Use the First Law.

Extension 4.19
An apple falls from a tree. The Earth pulls the apple down with a force of 1 N.
(a) What is the reaction force? Give its size and direction, and say what it acts on.
(b) The two forces are the same size. So why do we see the apple move towards the Earth, and not the Earth move towards the apple? Use the idea of inertia.

Extension 4.20
Two skaters stand still on ice and push each other apart. Skater A has twice the mass of skater B.
(a) Compare the sizes of the forces on the two skaters.
(b) Which skater moves away faster? Explain, using the idea of inertia.

Extension 4.21
A waiter carries a full cup of water on a tray.
(a) The waiter suddenly moves the tray forwards. Which way does the water spill? Explain.
(b) Later, walking at a steady speed, the waiter suddenly stops. Which way does the water spill now? Explain.
(c) Use the First Law to explain why moving the tray smoothly stops the water spilling.

5 Newton's Second Law: $F = ma$

Acceleration • Force, mass and acceleration • Rearranging $F = ma$ • Using the net force

Victorian Curriculum 2.0: **VC2S8U14 • VC2S10U17 (above level)**

Part A • Reading and comprehension

Reading — Pushing a Broken-Down Car

A car has broken down on a quiet, flat street, and it needs to be pushed to the side of the road. One person leans against the back and pushes as hard as they can. Slowly — very slowly — the car starts to roll. Two friends run over to help, and with three people pushing, the car speeds up much more quickly. A bigger force gives a bigger acceleration.

Further down the street, a removalist's van packed full of furniture has also broken down. The same three people push it just as hard as they pushed the car. This time it barely creeps forwards. The force is the same, but the van has far more mass, and more mass means a smaller acceleration.

Once the car is rolling, something changes: the pushers can ease off. They only need to push hard enough to match the friction acting on the car, and it keeps rolling at a steady speed. Pushing harder only matters if they want it to go faster.

Isaac Newton tied all of this together in his Second Law: how quickly an object speeds up depends on the net force acting on it, and on its mass. That is why racing cars and racing bikes are built to be as light as possible. With the same engine, or the same legs, a lighter vehicle speeds up faster.

C1. What happened to the car's acceleration when two friends joined in? Explain why.

C2. The same push was used on the loaded van. Describe what happened, and explain it using the word *mass*.

C3. Once the car rolls at a steady speed, which force are the pushers balancing? How do you know the forces are balanced?

C4. Why are racing cars and racing bikes built to be as light as possible?

C5. According to the passage, what *two* things decide how quickly an object speeds up?

C6. Which sentence in the first paragraph sums up what happened when more people pushed the car?

C7. Describe one other everyday example where the same push gives a smaller acceleration because the object has more mass.

C8. Suppose the three pushers kept pushing just as hard after the car was already rolling. Using the passage, predict what would happen to the car, and explain why using the words *net force*.

Part B • Theory and examples

5.1 What is Acceleration?

Acceleration is how quickly an object's speed changes. It is measured in **metres per second squared (m/s^2)**, which means "metres per second, every second".

Reading an acceleration

An acceleration of **3 m/s^2** means the speed goes up by **3 m/s every second**. Starting from rest:

Time (s)	0	1	2	3
Speed (m/s)	0	3	6	9

Slowing down is also an acceleration. It is sometimes called a **deceleration**.

5.2 Force, Mass and Acceleration

Newton's Second Law links three things: the **net force** on an object, its **mass**, and its **acceleration**.

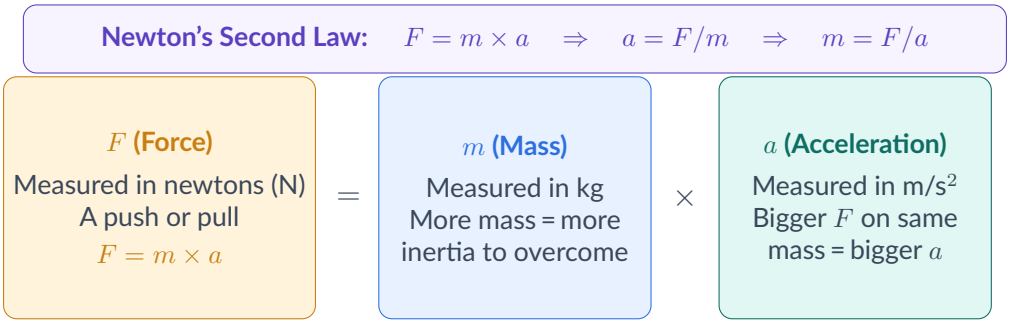


Figure 5.1 — Newton's Second Law: force, mass and acceleration

If you…	the acceleration…	Example
increase the force (same mass)	increases	Three people push the car instead of one
increase the mass (same force)	decreases	The same push on a loaded van
double the force (same mass)	doubles	Two people push a trolley, each as hard as one did before
double the mass (same force)	halves	The same push on two identical trolleys joined together

Second Law

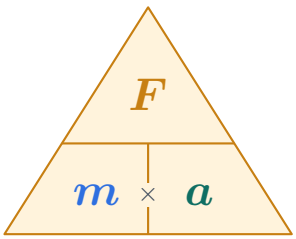
The acceleration of an object is directly proportional to the net force acting on it, and inversely proportional to its mass.

$$\boxed{F = m \times a} \qquad \boxed{a = \frac{F}{m}} \qquad \boxed{m = \frac{F}{a}}$$

Force in newtons (N) Mass in kilograms (kg) Acceleration in m/s^2

5.3 Using $F = ma$

You only need to remember one formula. The **formula triangle** shows how to rearrange it.



Cover F: $F = m \times a$

Cover a: $a = F \div m$

Cover m: $m = F \div a$

Side by side means multiply. One above the other means divide.

Figure 5.2 — The formula triangle. Cover the quantity you want; what is left shows how to find it.

Four steps for every $F = ma$ question

1. Write down what you **know**, with units.
2. Choose the **version** of the formula you need (use the triangle).
3. **Substitute** the numbers.
4. **Calculate**, and always write the **unit**.

Worked Example 5.1 — Finding acceleration

A 5 kg trolley is pushed with a net force of 20 N. What is its acceleration?

Know: $F = 20$ N, $m = 5$ kg
Formula: $F = m \times a \quad \Rightarrow \quad a = F \div m$
Substitute: $a = 20 \div 5 =$ **4 m/s^2**

Worked Example 5.2 — Finding force

A 1200 kg car accelerates at 2 m/s^2. What net force is acting on it?

Know: $m = 1200$ kg, $a = 2$ m/s^2
Formula: $F = m \times a$
Substitute: $F = 1200 \times 2 =$ **2400 N**

Worked Example 5.3 — Finding mass

A net force of 60 N gives a box an acceleration of 4 m/s^2. What is the mass of the box?

Know: $F = 60$ N, $a = 4$ m/s^2
Formula: $F = m \times a \quad \Rightarrow \quad m = F \div a$
Substitute: $m = 60 \div 4 =$ **15 kg**

5.4 Always Use the Net Force

The F in $F = ma$ is the **net force** — the force left over after you add up all the forces (Lesson 2). If the forces are balanced, the net force is zero, so the acceleration is zero: the object stays still, or keeps moving at a steady speed. That is exactly what the First Law says.

Worked Example 5.4 — Net force first

A 10 kg trolley is pushed forwards with 50 N. Friction pushes back with 10 N. Find its acceleration.

Step 1 — net force: opposite directions, so subtract: $50 - 10 = 40$ N forwards
Step 2 — acceleration: $a = F \div m = 40 \div 10 =$ **4 m/s^2 forwards**

Part C • Practice

Work through every task. Show your working for calculations, and always include the unit.

Task 5.1

For each change, describe what happens to the acceleration. Write **increases**, **decreases**, **doubles** or **halves**.

Change	Acceleration	Change	Acceleration
The force is doubled; the mass stays the same	*doubles*	The mass is doubled; the force stays the same	*halves*
More people help push the same car		A heavier rider pedals the same bike just as hard	
A truck is loaded with bricks; its engine force stays the same		A rocket burns fuel and gets lighter; its thrust stays the same	

Task 5.2

Use $F = ma$ to complete the table. The first row is done for you.

Situation	Force (N)	Mass (kg)	Acceleration (m/s^2)
Trolley pushed	20	5	$a = 20 \div 5 = 4$
Soccer ball kicked	45	0.45	
Car braking		1200	3.5
Rocket launch	6 000		30

Task 5.3

A net force of 24 N acts on a 6 kg mass. Find the acceleration.

Task 5.4

A 0.5 kg ball accelerates at 12 m/s^2. Find the net force.

Task 5.5
A force of 90 N gives an object an acceleration of 3 m/s^2. Find its mass.

Task 5.6
A 1500 kg car accelerates at 2.5 m/s^2. Find the net force needed.

Task 5.7
A 70 kg sprinter accelerates at 4 m/s^2 out of the blocks. Find the net force on her.

Task 5.8
A 0.15 kg cricket ball is hit with a net force of 30 N. Find its acceleration.

Task 5.9
A 2.5 kg trolley is pushed with different net forces.

(a) Complete the table.

Net force (N)	Mass (kg)	Acceleration (m/s^2)
5	2.5	$a = 5 \div 2.5 = 2$
10	2.5	
15	2.5	
20	2.5	

(b) What happens to the acceleration each time the force goes up by 5 N? Describe the pattern.

Task 5.10

A shopping trolley of mass 15 kg is pushed with a force of 45 N. Friction opposes the motion with 15 N.

(a) Find the net force.

(b) Find the acceleration of the trolley.

Task 5.11

A 60 kg cyclist and her 12 kg bicycle accelerate at 1.5 m/s^2.

(a) Find the total mass.

(b) Find the net force.

(c) Air resistance is 30 N. Find the force she must produce by pedalling.

Task 5.12

Two identical trolleys are pushed with the same force. Trolley A is empty, with a mass of 2 kg. Trolley B carries an extra 6 kg, so its total mass is 8 kg.

(a) Which trolley accelerates more?

(b) Explain your answer using $F = ma$.

(c) The force is 16 N. Calculate both accelerations.

Task 5.13

A net force of 3600 N gives a car an acceleration of 2 m/s^2.

(a) Find the mass of the car.

(b) The car is then loaded with 200 kg of luggage. Find its new acceleration with the same net force.

Task 5.14

A 900 kg car travels along a flat road at a steady 60 km/h. Its engine provides a driving force of 2000 N.

(a) What is the car's acceleration? Explain how you know.

(b) What is the total size of the resistive forces (friction and air resistance) acting on the car?

Part D • Above level

These questions go beyond the Year 7 standard. They ask you to combine ideas, use more than one formula, or judge someone else's reasoning.

Extension 5.15
Ahmed pushes a 20 kg crate with a force of 60 N, but it does not move. He says: *"$F = ma$ means it should accelerate at 3 m/s^2, so the formula must be wrong."* Explain Ahmed's mistake, and find the size of the friction force on the crate.

Extension 5.16
Mia pushes a 50 kg crate across the floor. Friction on the crate is 40 N. She says: *"If I push twice as hard, the crate will speed up twice as fast."*
(a) Find the acceleration when she pushes with 100 N.
(b) Find the acceleration when she pushes with 200 N.
(c) Is Mia right? Explain, using the idea of net force.

Extension 5.17
A rocket's engines push exhaust gas downwards with a force of 500 000 N, and the gas pushes the rocket upwards with the same force. The rocket has a mass of 20 000 kg. Use $g = 9.8$ N/kg.
(a) Find the rocket's weight.
(b) Find the net force on the rocket at lift-off.
(c) Find its acceleration.
(d) Which of Newton's laws explains (i) the upward push of the gas, and (ii) the size of the acceleration?

Extension 5.18
A lift and its passengers have a total mass of 800 kg. The cable pulls the lift upwards with a force of 8640 N. Use $g = 9.8$ N/kg.
(a) Find the weight of the lift and its passengers.
(b) Find the size and direction of the net force.
(c) Find the acceleration of the lift.

Extension 5.19
A cyclist speeds up from 2 m/s to 8 m/s in 4 s. The cyclist and bicycle have a total mass of 80 kg.
(a) Calculate the acceleration, using $a = \Delta v \div t$ (change in speed ÷ time).
(b) Calculate the net force on the cyclist and bicycle.
(c) Air resistance and friction total 40 N. Find the force the cyclist must supply by pedalling.

Extension 5.20
A 0.045 kg golf ball is struck and leaves the tee at 60 m/s. The club is in contact with the ball for 0.0005 s.
(a) Calculate the acceleration of the ball, using $a = \Delta v \div t$ (change in speed ÷ time).
(b) Calculate the force applied by the club.
(c) Calculate the ball's weight, and comment on how the force from the club compares with it.

Extension 5.21
In the trebuchet investigation at the end of this unit, a sling swings a 0.05 kg projectile. While it is being swung, the net force on the projectile is 2 N.
(a) Find the projectile's acceleration.
(b) The sling pulls on the projectile. Describe the reaction force.
(c) Once the projectile leaves the sling, nothing pushes it forwards. Explain why it keeps moving forwards, and name the force that brings it back to the ground.

6 Experiment 2: Newton's Second Law

Prediction • Variables • Trolley and pulley • Patterns in results

Victorian Curriculum 2.0: **VC2S8U14 • VC2S10U17 • VC2S8I02 • VC2S8I03 • VC2S8I04 • VC2S8I05**

Part A • Reading and comprehension

Reading — Slowing Down Gravity

Drop a ball and it hits the floor in well under a second — far too fast to measure by eye. Around 400 years ago, the Italian scientist Galileo Galilei faced exactly this problem. He wanted to find out how falling objects speed up, but he had no stopwatch, no camera and no clock accurate enough to help.

His clever solution was to slow everything down. Instead of dropping a ball, he let it roll down a long, gently sloping wooden ramp with a smooth groove cut into it. On a gentle slope, only a small part of the pull of gravity drives the ball along, so it speeds up slowly enough to measure. To time each roll, Galileo let water flow out of a large container through a thin pipe while the ball rolled, then weighed the water he had collected. More water meant more time.

Galileo's measurements showed that the ball sped up by the same amount every second — it had a steady acceleration. Isaac Newton, born less than a year after Galileo died, built on this work to explain how force, mass and acceleration are linked.

In this experiment you will use the same trick. A trolley rolls along a flat runway, pulled by a small hanging mass on a string. Only the weight of that small mass pulls the trolley along, so it speeds up slowly enough for you to measure. By changing the hanging mass, you change the pulling force — and you can test Newton's Second Law for yourself.

C1. Why couldn't Galileo simply time a ball falling straight down?

C2. Explain how a gentle ramp helped Galileo, and how he measured time without a clock.

C3. In this experiment, what provides the force that pulls the trolley along?

C4. Why does using a *small* hanging mass make the trolley's motion easier to measure?

Part B • Planning the experiment

Experiment 2: Investigating Newton's Second Law

Curriculum: VC2S8U14 • VC2S10U17 • VC2S8I02 • VC2S8I03 • VC2S8I04 • VC2S8I05

Aim: *To investigate how the size of the pulling force affects the acceleration of a trolley, when the total mass stays the same.*

6.1 The Set-up

A trolley on a flat runway is joined by a string, over a pulley, to a hanging mass. The weight of the hanging mass provides the pulling force. The pulley simply turns the direction of the pull from *down* to *sideways*.

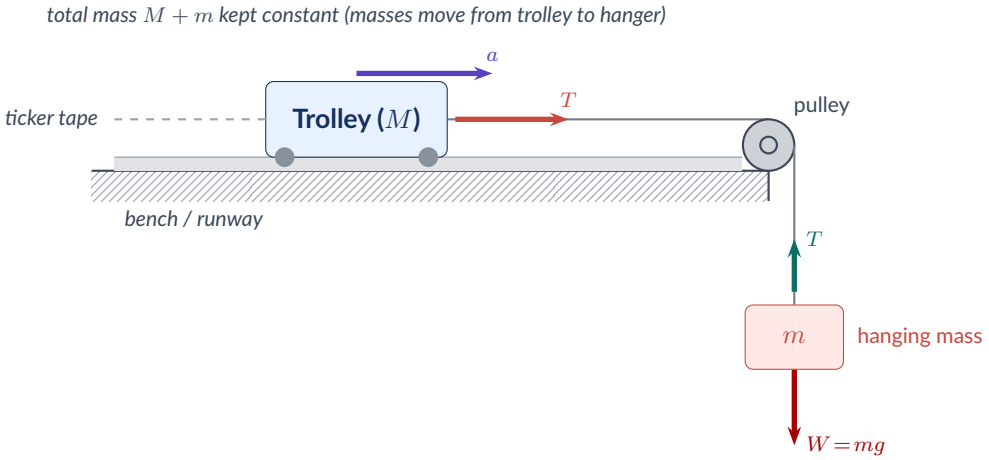


Figure E6.1 — Trolley connected by string over a pulley to a hanging mass. The hanging mass provides the driving force; the pulley redirects the tension.

6.2 The Forces on the Trolley

Vertically, the trolley's weight and the normal force from the runway are **balanced**. Horizontally, the tension in the string is the only force that matters (friction is very small), so it is **unbalanced** — and the trolley accelerates.

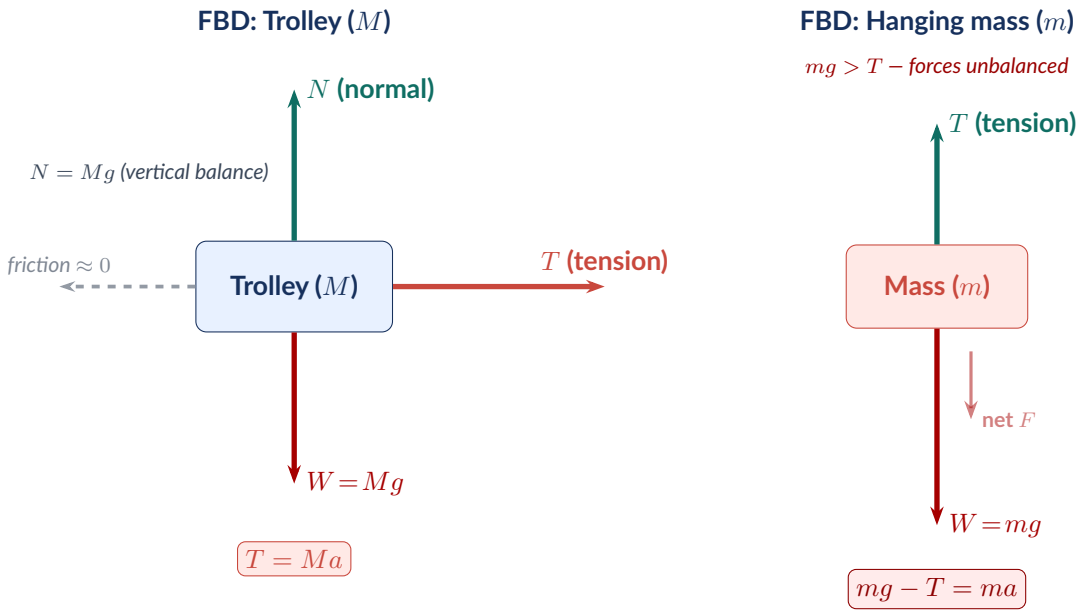


Figure E6.2 — Left: free body diagram of the trolley (tension drives it right; normal force and weight balance vertically). Right: free body diagram of the hanging mass (weight is larger than tension, so it accelerates downwards).

6.3 Calculating the Pulling Force

The pulling force is the weight of the hanging mass, so use $W = mg$ with $g = 9.8$ N/kg. The mass must be in **kilograms**: divide grams by 1000.

Worked Example 6.1 — Finding the pulling force

A 150 g mass hangs on the end of the string. Find the pulling force on the trolley.

Step 1 — convert to kg: $150 \div 1000 = 0.150$ kg
Step 2 — weight: $W = mg = 0.150 \times 9.8 = \mathbf{1.47}$ N

6.4 What $F = ma$ Predicts

If $F = ma$, then doubling the force on the same mass should double the acceleration. This experiment tests that prediction.

Prediction — before you begin

What do you think will happen to the trolley's acceleration as the hanging mass is increased from 50 g to 200 g? Use $F = ma$ to give a reason.

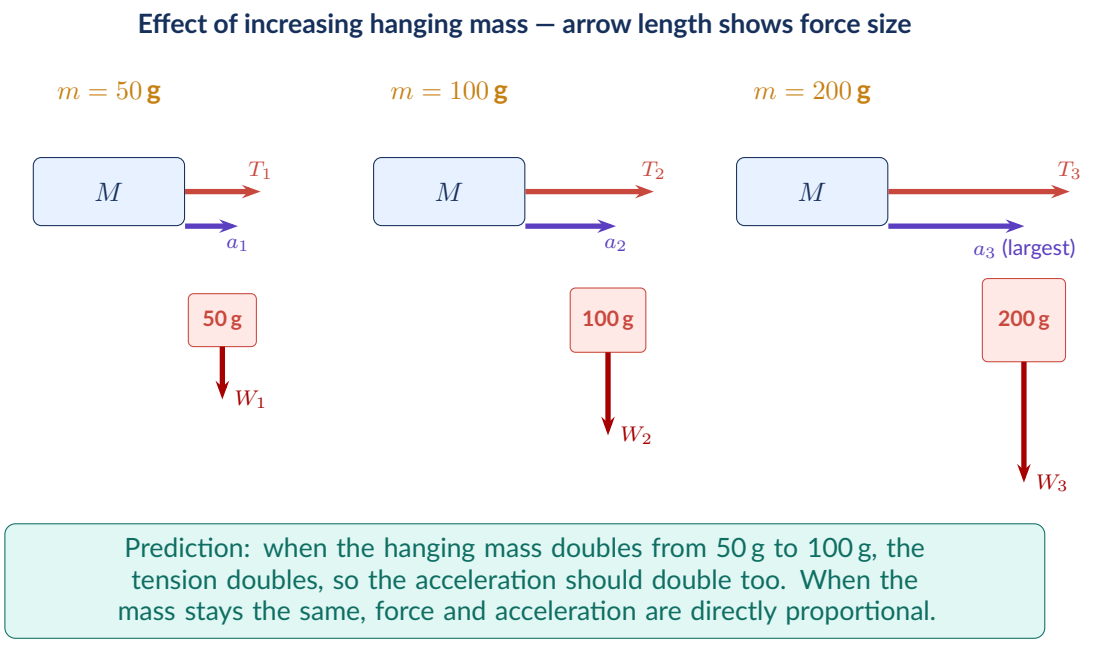


Figure E6.3 — Check your prediction against what Newton's Second Law predicts: as the hanging mass increases, the tension force increases, and so does the acceleration.

Equipment

Dynamics trolley Runway Pulley (clamped to the end of the bench) and string Mass hanger and slotted masses (4 × 50 g) Motion sensor or ticker-tape timer Balance Buffer block and catch box

Method

1. Measure the mass of the trolley with the balance and record it.
2. Clamp the pulley to the end of the bench. Tie a string from the trolley, over the pulley, to the mass hanger.
3. Load the hanger so the hanging mass is **50 g**. Put the other three 50 g masses **on the trolley**.
4. Hold the trolley still at the start line, then let go — do not push it. Measure its acceleration with the motion sensor or ticker-tape timer.
5. Repeat 3 times and record every result.
6. Move one 50 g mass from the trolley to the hanger (the hanging mass is now 100 g) and repeat steps 4–5. Do the same for 150 g and 200 g.

Why move the masses instead of adding new ones?

Moving a mass from the trolley to the hanger keeps the **total mass** being pulled the same. Only the **force** changes — so it is a fair test.

Safety

Put a buffer block at the end of the runway and catch the trolley before it hits the pulley. Place a catch box under the hanging mass and keep your feet clear of it. Never overload the hanger.

Part C • Results and analysis

Results

Hanging mass (g)	Pulling force $F = mg$ (N)	Trial 1 (m/s^2)	Trial 2 (m/s^2)	Trial 3 (m/s^2)	Mean accel. (m/s^2)
e.g. 50	*0.49*	*0.38*	*0.42*	*0.41*	*0.40*
50					
100					
150					
200					

Looking for a pattern

Two quantities are **directly proportional** if doubling one doubles the other. To check, compare two rows: when the force doubles (50 g → 100 g), does the mean acceleration roughly double too?

Task 6.1

Identify the independent variable, the dependent variable, and two controlled variables in this experiment.

Task 6.2

Why were the 50 g masses moved from the trolley to the hanger, instead of adding new masses to the hanger?

Task 6.3

Describe the relationship between force and acceleration that you observed. Did doubling the force (50 g $\rightarrow$ 100 g) roughly double the acceleration? Was your prediction correct?

Task 6.4

A different hanging mass of 120 g is used. Find the pulling force on the trolley.

Task 6.5

Another group measured 0.81, 0.79 and 0.95 m/s^2 for 100 g. Which value looks wrong? Find the mean of the other two.

Task 6.6

Use Newton's Second Law to explain *why* a larger hanging mass gave the trolley a larger acceleration. Use the words *net force* and *mass*.

Task 6.7

Draw a line graph of your results. Put **pulling force (N)** on the horizontal axis and **mean acceleration (m/s^2)** on the vertical axis. Plot each point, draw a line of best fit, and give your graph a title.

Part D • Above level

These questions go beyond the Year 7 standard. They ask you to interpret your graph, use a formula, or evaluate and extend the method.

Extension 6.8
Look at your graph from Task 6.7.
(a) Is your line of best fit straight, and does it pass through (or close to) the origin? What does this tell you about force and acceleration?
(b) Calculate the gradient of your line. What does it represent?

Extension 6.9
For each row of your results, calculate $F \div a$ (pulling force ÷ mean acceleration).
(a) What do you notice about your answers?
(b) Use $F = ma$ to explain what $F \div a$ should be equal to.
(c) Compare your values with the total mass of the trolley plus all four 50 g masses.

Extension 6.10
Combining the two free body diagrams (Figure E6.2) gives a formula for the acceleration of the whole system:

System analysis: combining both free body diagrams

Trolley (M):
Horizontal net force $= T$
$\Rightarrow \quad T = Ma \quad \cdots (1)$

Hanging mass (m):
Net force $= mg - T$ downward
$\Rightarrow \quad mg - T = ma \quad \cdots (2)$

Add (1) and (2) to eliminate T:

$$mg = Ma + ma = (M + m)\, a$$

$$a = \frac{mg}{M + m}$$

When $M \gg m$, the total mass $\approx M$, so $a \approx \dfrac{mg}{M} = \dfrac{F}{M}$ — confirming $F = Ma$

Figure E6.4 — Combining both free body diagrams gives the full system equation. When $M \gg m$, the result simplifies to $F = Ma$.

In this experiment the total mass $M + m$ (the trolley plus all four slotted masses, wherever they sit) is 1.2 kg.
(a) Calculate the predicted acceleration when the hanging mass is 100 g.
(b) Your measured acceleration was probably a little smaller than this. Suggest one reason, and one change to the method that would reduce the difference.

Extension 6.11
Design a second fair test, to find out how the *mass* of the trolley affects its acceleration.
(a) Name the independent and dependent variables.
(b) What must you keep the same? Explain how you would keep the pulling force the same.
(c) Predict what you would find, and give a reason using $F = ma$.

7 Energy and Transformation

Forms of energy • Energy chains • Conservation and efficiency • GPE and KE

Victorian Curriculum 2.0: **VC2S8U15 • VC2S10U15 (above level)**

Part A • Reading and comprehension

Reading — Why the First Hill is Always the Tallest

Look carefully at a traditional roller coaster — the kind that is pulled up its first hill by a chain — and you will notice something: the first hill is always the highest one on the track. This is not a matter of taste or thrill design. It is a rule imposed by physics, and no engineer can get around it. The train itself has no engine. A chain drags it slowly up that first hill, and after that the ride is entirely powered by gravity. Hauling the train to the top stores energy in it — gravitational potential energy — and the amount stored depends on how high the train is lifted. As the train plunges down, that stored energy is converted into kinetic energy, the energy of movement, and the train reaches its greatest speed at the bottom. Climbing the next hill converts kinetic energy back into potential energy, and so the ride continues, trading one form for the other.

The catch is that the swap is never perfect. Every time the wheels turn, a little energy is lost as heat through friction, and a little more is lost as sound — that familiar roar. Because energy cannot be created, the train can never climb higher than it started. Each successive hill must be a little lower than the one before, and the ride must end before the energy runs out entirely.

At the end of the ride the train still has some kinetic energy, and it has to be removed safely. Many modern coasters use magnetic brakes: metal fins under the train pass between strong magnets, and the magnets slow the fins down without ever touching them. The train's kinetic energy is not destroyed — it ends up as thermal energy, warming the fins and the air around them.

C1. Where does a roller coaster train get the energy for the whole ride, and how is that energy stored?

C2. Describe the energy transformation that happens as the train travels down the first hill.

C3. Explain why each hill on the track must be lower than the one before it. Name the two ways energy is lost.

C4. At which point on the ride does the train have the most kinetic energy? Explain how you know.

C5. Which phrase in the third paragraph tells you that energy cannot simply appear from nowhere?

C6. How do magnetic brakes stop the train, and what happens to the train's kinetic energy?

C7. In your own words, explain what the passage means when it says the swap between potential and kinetic energy is "never perfect".

C8. Imagine the track and wheels were so smooth that there was almost no friction. Using the passage, predict how the heights of the later hills could change, and explain why.

Part B • Theory and examples

7.1 Forms of Energy

Energy is what makes things happen: it is needed to move something, heat it, light it up or make a sound. Energy is measured in **joules (J)**. It comes in several forms. Some are *stored* energy, waiting to be used; others show that something is *happening*.

Form of energy	What it is	Example
Kinetic	The energy of anything that is moving	A rolling ball, a moving car
Gravitational potential	Energy stored by lifting something up high	A book on a high shelf
Elastic potential	Energy stored by stretching or squashing	A stretched rubber band
Chemical	Energy stored in food, fuel and batteries	A sandwich, petrol, a battery
Thermal	Energy of hot objects; hotter means more	A cup of hot tea
Sound, light and electrical	Energy carried by sound waves, light and electric current	A speaker, a torch beam, a power cable

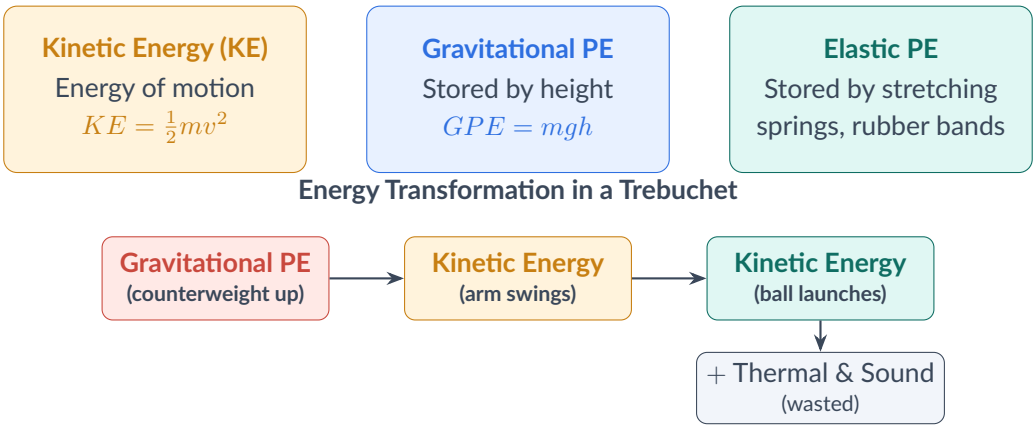


Figure 7.1 — Three forms of energy you will meet again and again, and how energy changes form in a trebuchet

7.2 Transformations and Energy Chains

Energy can change from one form to another. This is called an **energy transformation**. An **energy chain** shows each step, with an arrow for every change. The trebuchet in Figure 7.1 turns the gravitational potential energy of its raised counterweight into kinetic energy of the arm, and then of the ball.

Worked Example 7.1 — An energy chain for a bouncing ball

A ball is held above the floor, dropped, and bounces back up. Write the energy chain from the moment it is let go to the top of its bounce.

Falling: gravitational potential → kinetic (it speeds up as it falls)
Hitting the floor: kinetic → elastic potential (the ball squashes)
Bouncing up: elastic potential → kinetic → gravitational potential

At every step a little energy is **wasted** as thermal energy and sound — you can hear the bounce.

7.3 Conservation of Energy and Efficiency

The Law of Conservation of Energy

Energy **cannot be created or destroyed**. It can only be transformed from one form to another, or transferred from one object to another. The total amount of energy at the end is always the same as at the start.

So where does energy "go"? Not all of it ends up where we want it. The energy that does the job we want is **useful** energy. The rest is **wasted** energy — usually thermal energy and sound, which spread out into the surroundings and are hard to use again. A **Sankey diagram** shows this with arrows whose width matches the amount of energy.

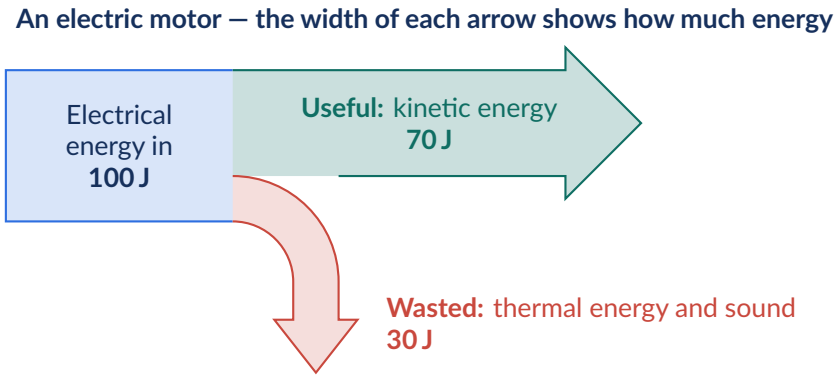


Figure 7.2 — A Sankey diagram. The useful and wasted arrows always add up to the energy in.

Efficiency

$$\text{Efficiency} = \frac{\text{useful energy out}}{\text{total energy in}} \times 100\%$$

A machine that wastes no energy would be 100% efficient. No real machine is.

Worked Example 7.2 — Calculating efficiency

An old-style light bulb is supplied with 100 J of electrical energy. It gives out 5 J of light. How much energy is wasted, and what is its efficiency?

Wasted: $100 - 5 = \mathbf{95}$ J (as thermal energy — the bulb gets hot)
Efficiency: $5 \div 100 \times 100\% = \mathbf{5\%}$

7.4 Calculating Gravitational Potential and Kinetic Energy

Two energy formulas

$$\text{GPE} = m \times g \times h \qquad \text{KE} = \tfrac{1}{2} \times m \times v^2$$

m = mass (kg) $g = 9.8$ N/kg on Earth h = height (m) v = speed (m/s) Energy in joules (J)

Worked Example 7.3 — Gravitational potential energy

A 2 kg ball is held at a height of 5 m. Find its gravitational potential energy.

GPE $= mgh = 2 \times 9.8 \times 5 =$ **98** J

Worked Example 7.4 — Kinetic energy

A 4 kg dog runs at 3 m/s. Find its kinetic energy.

Square the speed first: $v^2 = 3 \times 3 = 9$
Then: KE $= \frac{1}{2} \times m \times v^2 = 0.5 \times 4 \times 9 =$ **18** J

When the 2 kg ball in Worked Example 7.3 is dropped, its 98 J of gravitational potential energy becomes kinetic energy. Just before it lands it has close to 98 J of kinetic energy — a little less, because some is wasted pushing through the air.

Notes

Part C • Practice

Work through every task. Use $g = 9.8$ N/kg. Show your working for calculations, and always include the unit.

Task 7.1

For each object, name the main form of energy it has. Write it in the box.

Object	Form of energy	Object	Form of energy
A stretched rubber band	*elastic potential*	A bowling ball rolling down the lane	*kinetic*
A rock balanced on a cliff edge		A battery	
A cup of hot chocolate		A drawn-back bowstring	
A sandwich		A child sliding down a slide	

Task 7.2

Complete the table. For each device, name the form of energy that goes in, the useful energy that comes out, and the main wasted energy.

Device	Energy in	Useful energy out	Wasted energy
Torch	*chemical (battery)*	*light*	*thermal*
Electric kettle			
Car engine			
Wind-up toy car			

Task 7.3

A person bounces on a trampoline. Write the energy chain from the top of one jump, down onto the trampoline, and back up to the top of the next jump. Then name the energy that is wasted.

Task 7.4
A 2 kg book is lifted onto a shelf 1.8 m high. Find its GPE.

Task 7.5
A 0.15 kg ball moves at 20 m/s. Find its KE.

Task 7.6
A 60 kg student climbs stairs 4 m high. Find the GPE gained.

Task 7.7
A 1200 kg car travels at 15 m/s. Find its KE.

Task 7.8
A 3 kg ball is held at a height of 4 m.

(a) Calculate its GPE.

(b) The ball is dropped. Just before it hits the ground its speed is 8.85 m/s. Calculate its KE.

(c) Compare your answers to (a) and (b). What does this suggest?

Task 7.9
Complete the table. Use efficiency $=$ useful energy out $\div$ energy in $\times\ 100\%$.

Device	Energy in (J)	Useful energy out (J)	Wasted energy (J)	Efficiency (%)
Old-style light bulb	100	5	$100 - 5 = 95$	$5 \div 100 \times 100 = 5$
Electric motor	200	150		
Electric kettle	2000	1800		
Car engine	500	150		

Task 7.10

A machine is supplied with 800 J of energy and produces 560 J of useful output.

(a) Calculate the efficiency.

(b) How much energy was wasted?

(c) Suggest two forms the wasted energy is likely to take.

Task 7.11

Look at the trebuchet energy chain in Figure 7.1.

(a) Name **two** forms of energy present at the moment the trebuchet releases the ball.

(b) Explain why the ball's kinetic energy is less than the gravitational potential energy the counterweight started with.

Task 7.12

A 0.5 kg ball is dropped from a height of 3 m.

(a) Calculate its GPE before it is dropped.

(b) Assuming no energy is wasted, state its KE just before it lands.

(c) In real life, would its KE be a little more or a little less than your answer to (b)? Explain.

Task 7.13

A pendulum is pulled to one side and let go. It swings back and forth.

(a) Where in the swing does it have the most gravitational potential energy?

(b) Where does it have the most kinetic energy?

(c) Explain why it eventually stops swinging.

Part D • Above level

These questions go beyond the Year 7 standard. They ask you to use the Law of Conservation of Energy, combine formulas, or judge someone else's reasoning.

Extension 7.14
The 2 kg ball in Worked Example 7.3 starts with 98 J of GPE, but actually hits the ground with 92 J of KE.
(a) Calculate the efficiency of the GPE $\rightarrow$ KE transformation.
(b) Explain where the "missing" energy went. Has it been destroyed?

Extension 7.15
A trebuchet counterweight provides 500 J of GPE. Only 380 J is transferred to the ball as KE. Draw a Sankey diagram for this energy transfer, using a scale of **1 cm = 50 J**. Label each arrow with its energy.

Draw your Sankey diagram here: energy in on the left, useful energy straight on, wasted energy turning down.

Extension 7.16
A 0.5 kg ball is dropped from a height of 3 m. Assume no energy is wasted.
(a) Find its KE just before it lands.
(b) Use KE $= \frac{1}{2}mv^2$ to find the speed at which it lands. *(Hint: find v^2 first, then take the square root.)*

Extension 7.17
A trebuchet's 10 kg counterweight falls 1.0 m. The trebuchet is 20% efficient.
(a) Calculate the GPE given up by the counterweight.
(b) Calculate the KE given to the projectile.
(c) The projectile has a mass of 0.1 kg. Calculate its launch speed.

Extension 7.18
Jordan says: *"If a car goes twice as fast, it has twice as much kinetic energy."*
(a) Calculate the KE of a 1000 kg car at 10 m/s and at 20 m/s.
(b) Is Jordan right? Explain what happens to the KE when the speed doubles.
(c) Suggest why this matters for how far a car needs to stop.

Extension 7.19
A 500 kg roller coaster train is pulled to the top of a first hill 40 m high.
(a) Calculate its GPE at the top.
(b) By the bottom of the first drop, 25% of this energy has been wasted. Calculate its KE at the bottom.
(c) Use your answers to explain why the next hill cannot be 40 m high.

Extension 7.20
A pole vaulter sprints down the runway, plants the pole, bends it, and is flung up over the bar. Describe the energy chain from the run-up to the top of the jump. Say where elastic potential energy appears, and where energy is wasted.

8 Experiment 3: Bounce Efficiency

Prediction • Drop height and bounce height • Repeat trials • Calculating efficiency

Victorian Curriculum 2.0: **VC2S8U15 • VC2S8I01 • VC2S8I02 • VC2S8I03 • VC2S8I04 • VC2S8I05**

Part A • Reading and comprehension

Reading — The Rules of the Bounce

Every ball used in serious sport has to pass a bounce test. A tennis ball, for example, is dropped from a height of 254 centimetres onto concrete. To be approved for play, it must bounce back up to between 135 and 147 centimetres — a little more than half the height it fell from.

Why doesn't a ball bounce all the way back up? As it falls, its gravitational potential energy becomes kinetic energy. When it hits the ground the ball squashes, and for a split second its energy is stored as elastic potential energy, like a squashed spring. The ball then springs back into shape and pushes itself off the ground. But the rubber does not spring back perfectly: some energy becomes thermal energy inside the rubber, and some becomes the sound of the bounce. Less energy comes out than went in, so the ball cannot rise as high as it fell.

Squash players know this well. A cold squash ball hardly bounces at all, so players hit it hard for a few minutes before a match. As the ball warms up, the air inside it pushes harder on the rubber, and the rubber itself becomes springier, so the ball bounces higher. The ball used by the best players in the world bounces the least of all, which makes it the hardest to keep in play.

In this experiment you will run your own bounce test, and use the heights to work out how efficient each bounce is.

C1. From what height is a tennis ball dropped in the bounce test, and how high must it bounce to be approved?

C2. Explain, using the words *elastic potential energy*, what happens to the ball's energy when it hits the ground.

C3. Why can a ball never bounce back up to the height it was dropped from? Name the two forms of wasted energy.

C4. Why do squash players hit the ball hard for a few minutes before a match?

Part B • Planning the experiment

Experiment 3: Bounce Efficiency

Curriculum: VC2S8U15 • VC2S8I01 • VC2S8I02 • VC2S8I03 • VC2S8I04 • VC2S8I05

Aim: *To investigate how the drop height of a ball affects its bounce height, and to calculate the efficiency of each bounce.*

8.1 Measuring a Bounce

When the ball is held still at the drop height, all of its useful energy is gravitational potential energy. At the top of the bounce it is still again, so all of its energy is gravitational potential energy once more. Because GPE $= mgh$, and the mass and g do not change, comparing the two *heights* tells you how much of the energy was kept.

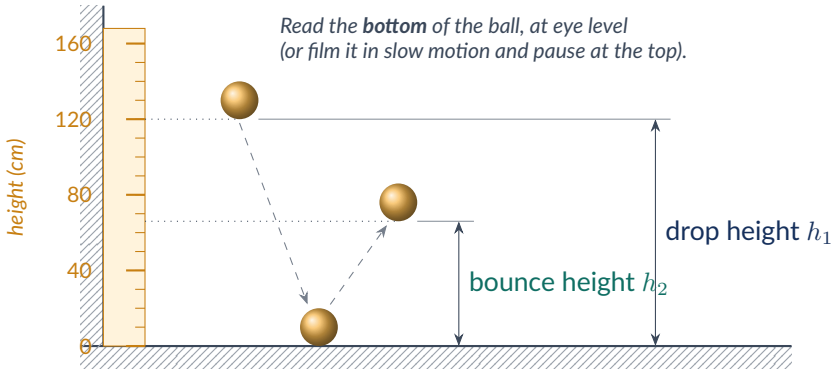


Figure E8.1 — Measuring the drop height h_1 and the bounce height h_2 against metre rulers fixed to a wall.

Bounce efficiency

$$\text{Efficiency} = \frac{\text{bounce height } h_2}{\text{drop height } h_1} \times 100\%$$

Worked Example 8.1 — Calculating bounce efficiency

A ball is dropped from 100 cm and bounces back up to 62 cm. Find the efficiency of the bounce, and the percentage of energy wasted.

Efficiency: $62 \div 100 \times 100\% = \mathbf{62\%}$
Wasted: $100\% - 62\% = \mathbf{38\%}$ (as thermal energy and sound)

8.2 Making it a Fair Test

You will change the **drop height** (independent variable) and measure the **bounce height** (dependent variable). Everything else must stay the same: the same ball, the same patch of floor, letting go without pushing, and measuring to the *bottom* of the ball every time.

Prediction — before you begin

What do you think will happen to the bounce height as the drop height increases from 40 cm to 160 cm? Will the *efficiency* change? Give a reason.

Equipment

Tennis ball (or another rubber ball) Two metre rulers, or a tape measure, fixed upright to a wall Sticky tape Hard, flat floor Phone with slow-motion video (optional)

Method

1. Fix the metre rulers upright against the wall, with 0 cm touching the floor.
2. Hold the ball beside the rulers so that the **bottom** of the ball is level with the 40 cm mark.
3. Let go — do not push or throw the ball.
4. Watch (or film) the bounce, and read the height of the bottom of the ball at the top of its first bounce. Record it.
5. Repeat steps 2–4 so that you have 3 trials.
6. Repeat for drop heights of 80 cm, 120 cm and 160 cm.

Getting a good reading

The top of the bounce lasts only a moment. Crouch so your eyes are level with where the ball will stop, or film the bounce in slow motion and pause it at the highest point. One person drops, one person reads.

Safety

Keep every drop height within reach while you are standing on the floor — do not stand on chairs or desks. Clear the area of bags and chairs, and let the ball come to rest before you pick it up.

Notes

Part C • Results and analysis

Results

Drop height h_1 (cm)	Bounce, trial 1 (cm)	Bounce, trial 2 (cm)	Bounce, trial 3 (cm)	Mean bounce h_2 (cm)	Efficiency (%)
e.g. 40	*23*	*21*	*22*	*22*	*55*
40					
80					
120					
160					

Calculating the mean and the efficiency

Mean: add the three trials and divide by 3. For 23, 21 and 22 cm: $(23 + 21 + 22) \div 3 = \mathbf{22}$ cm.
Efficiency: $22 \div 40 \times 100\% = \mathbf{55}\%$.

Task 8.1

Identify the independent variable, the dependent variable, and two controlled variables in this experiment.

Task 8.2

What happened to the bounce height as the drop height increased? Was your prediction correct?

Task 8.3

Look at your efficiency column. Did the efficiency stay about the same, rise, or fall as the drop height increased? Use numbers from your results.

Task 8.4

A ball dropped from 150 cm bounces to 90 cm. Calculate the efficiency of the bounce.

Task 8.5

Another group's bounce heights for an 80 cm drop were 52, 55 and 41 cm. Which value looks wrong? Find the mean of the other two.

Task 8.6

Explain why the ball never bounces back up to its drop height. Use the words *gravitational potential*, *kinetic*, *elastic potential* and *thermal* in your answer.

Task 8.7

Draw a line graph of your results. Put **drop height (cm)** on the horizontal axis and **mean bounce height (cm)** on the vertical axis. Plot each point, draw a line of best fit, and give your graph a title.

Part D • Above level

These questions go beyond the Year 7 standard. They ask you to interpret your graph, use a formula, or evaluate and extend the method.

Extension 8.8
Look at your graph from Task 8.7.
(a) Is your line of best fit straight, and does it pass through (or close to) the origin? What does this tell you about drop height and bounce height?
(b) Calculate the gradient of your line. Explain how it is connected to the efficiency of the ball.

Extension 8.9
A 0.2 kg ball is dropped from 2.0 m and bounces back up to 1.4 m. Use $g = 9.8$ N/kg.
(a) Calculate the GPE before the drop, and at the top of the bounce.
(b) Calculate the efficiency of the bounce.
(c) How much energy was wasted, and what forms did it take?

Extension 8.10
Predict how the bounce efficiency would change if you dropped the same ball onto thick carpet instead of a hard floor. Explain your prediction using the ideas of elastic potential energy and thermal energy.

Extension 8.11
Design a fair test to find out whether the *temperature* of a ball changes its bounce efficiency.
(a) Name the independent and dependent variables.
(b) Name two variables you would keep the same.
(c) Describe a safe way to make the ball warmer and colder.
(d) Predict what you would find, and give a reason using the reading.

9 Simple Machines and Levers

Simple machines • Parts of a lever • Mechanical advantage • Lever classes • The trebuchet

Victorian Curriculum 2.0: **VC2S8U13 • VC2S8U14**

Part A • Reading and comprehension

Reading — Lifting Stones for a Pharaoh

The Great Pyramid of Giza contains roughly 2.3 million blocks of limestone. Many weigh more than two tonnes, and some of the granite slabs inside weigh as much as eighty. The pyramid was completed around 4,500 years ago, long before engines, hydraulics or steel cable existed. The builders had rope, timber, copper tools and their own muscles — and yet the structure stood as the tallest building on Earth for nearly four thousand years.

Historians think much of the answer lies in simple machines. A *lever* lets a small effort applied over a long distance produce a large force over a short one. A *ramp*, or inclined plane, does the same thing in a different way: dragging a block up a long gentle slope needs far less force than lifting it straight up, though you must move it much further. Neither device creates energy — that would be impossible — but both let human beings trade distance for force, which is exactly the trade a person with a rope and a lot of patience needs to make.

The same principle drives the machine you will build towards in this unit. A trebuchet is essentially a giant lever. A heavy counterweight falls a short distance on the short arm, and the long arm sweeps through a huge arc, flinging the projectile at high speed. The Egyptians used levers to move stone slowly and deliberately. Medieval engineers used the same physics to throw it as fast and as far as they possibly could.

C1. What tools and materials did the pyramid builders have available to them?

C2. Explain what is meant by “trade distance for force”. Use the ramp as your example.

C3. According to the passage, what do the pyramid builders’ levers and a medieval trebuchet have in common?

C4. About how many blocks are in the Great Pyramid, and how heavy are many of them?

C5. Which sentence in the second paragraph tells you that a simple machine cannot give you extra energy?

C6. In your own words, explain how the Egyptians and the medieval engineers used levers for *different* purposes.

C7. Name the two simple machines described in the passage, and give one other example of a lever you have used.

C8. Suppose the builders used a ramp twice as long to reach the same height. Using the passage, predict how the force needed and the distance moved would change.

Part B • Theory and examples

9.1 Simple Machines

A **simple machine** is a device that changes the *size* or the *direction* of a force, making a job easier. There are six of them.

Simple machine	How it helps	Example
Lever	A bar that turns on a pivot	Seesaw, crowbar, scissors
Inclined plane	A slope: less force, longer distance	Ramp, wheelchair ramp
Wedge	Two slopes back to back that push things apart	Axe, knife, doorstop
Wheel and axle	A large wheel turns a small axle	Doorknob, steering wheel
Pulley	A wheel and rope that change the direction of a pull	Flagpole, crane
Screw	An inclined plane wrapped around a rod	Jar lid, screw, bolt

Think about it — no free energy

A simple machine never gives you more energy than you put in. If it lets you push with **less force**, you must push through a **longer distance**. That is the trade every simple machine makes.

9.2 The Parts of a Lever

A **lever** is a rigid bar that turns about a fixed point called the **fulcrum** (or pivot). The **effort** is the force you apply; the **load** is the force the lever moves or lifts. The distance from the fulcrum to the effort is the **effort arm**; the distance from the fulcrum to the load is the **load arm**.

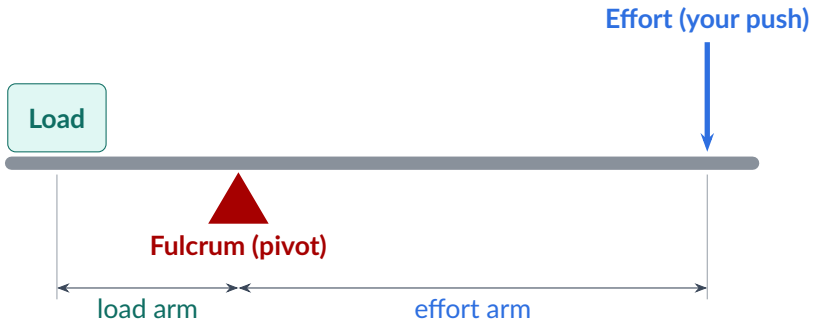


Figure 9.1 — The parts of a lever. A long effort arm lets a small effort lift a large load.

The law of the lever

A lever is balanced when effort $\times$ effort arm $=$ load $\times$ load arm.

Worked Example 9.1 — Balancing a seesaw

A 300 N child sits 2 m from the pivot of a seesaw. Where must a 400 N child sit on the other side to balance it?

Law of the lever: $300 \times 2 = 400 \times d$
Solve: $600 = 400 \times d$ so $d = 600 \div 400 =$ **1.5** m

The heavier child sits *closer* to the pivot.

9.3 Mechanical Advantage

Mechanical advantage (MA)

$$\text{MA} = \frac{\text{load}}{\text{effort}} = \frac{\text{effort arm}}{\text{load arm}}$$

MA bigger than 1: the lever **multiplies your force**. MA smaller than 1: you lose force but **gain speed and distance**.

Worked Example 9.2 — Mechanical advantage

A lever has an effort arm of 2.0 m and a load arm of 0.5 m. Find its MA, and the load that a 100 N effort can lift.

MA: $2.0 \div 0.5 = \mathbf{4}$
Load: load $=$ MA $\times$ effort $= 4 \times 100 = \mathbf{400}$ N

9.4 The Three Classes of Lever

Levers are sorted into three classes by which part is in the **middle**. A memory aid: **F–L–E, 1–2–3**: the **F**ulcrum is in the middle of a Class 1 lever, the **L**oad in the middle of Class 2, and the **E**ffort in the middle of Class 3.

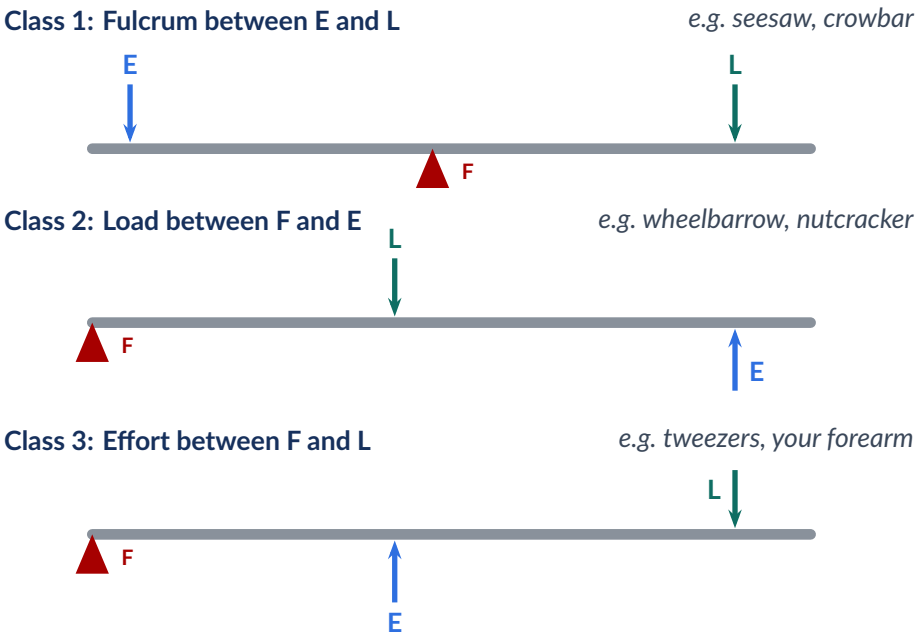


Figure 9.2 — The three classes of lever (F = Fulcrum, E = Effort, L = Load)

Class	In the middle	What it does	Examples
Class 1	Fulcrum	Can multiply force or speed	Seesaw, crowbar, scissors
Class 2	Load	Always multiplies force (MA $>$ 1)	Wheelbarrow, nutcracker, bottle opener
Class 3	Effort	Always multiplies speed (MA $<$ 1)	Tweezers, your forearm, a fishing rod

9.5 The Trebuchet as a Class 1 Lever

A trebuchet arm turns on a pivot between the counterweight and the projectile, so it is a **Class 1 lever**. The falling counterweight is the **effort**, on the *short* arm. The projectile is the **load**, at the end of the *long* arm. Its MA is therefore **less than 1**: it gives away force to gain speed. In the time the counterweight drops a short distance, the end of the long arm sweeps through a much longer arc — so it moves much faster.

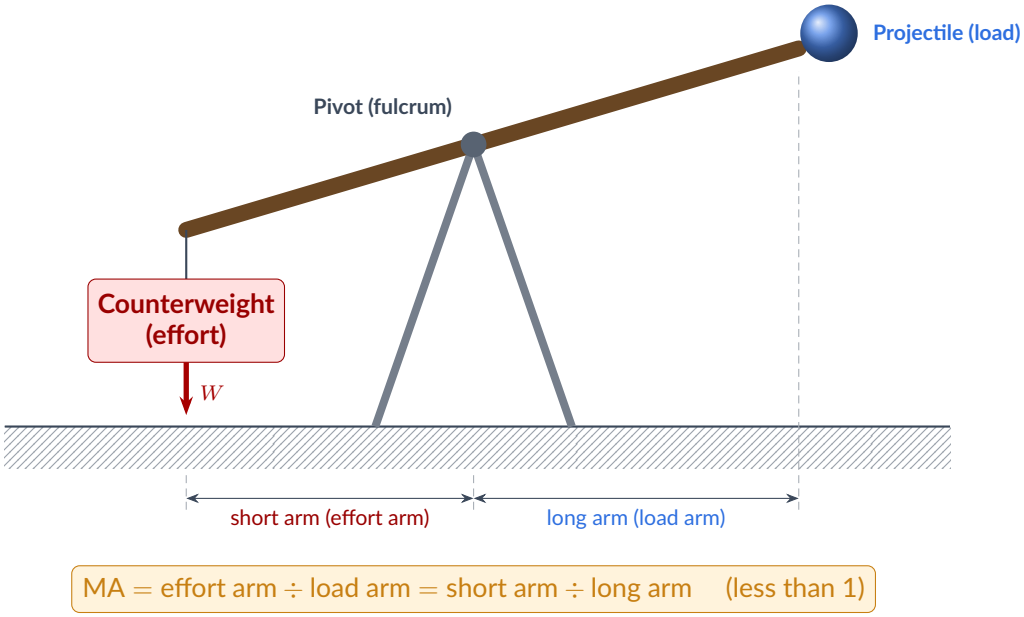


Figure 9.3 — A trebuchet is a Class 1 lever with a short effort arm and a long load arm.

Worked Example 9.3 — The trebuchet's mechanical advantage

A trebuchet has a short arm of 0.4 m and a long arm of 2.0 m. Find its MA, and how many times faster the end of the long arm moves than the counterweight.

MA: effort arm ÷ load arm $= 0.4 \div 2.0 =$ **0.2** (less than 1: force is given away)
Speed: long arm ÷ short arm $= 2.0 \div 0.4 =$ **5** so the arm tip moves **5 times** as fast as the counterweight.

Notes

Part C • Practice

Work through every task. Show your working for calculations, and always include the unit.

Task 9.1

Name the simple machine in each object. Write it in the box.

Object	Simple machine	Object	Simple machine
A ramp up into a removal truck	*inclined plane*	A crowbar	*lever*
An axe splitting wood		The rope and wheel on a flagpole	
The lid of a jam jar		A doorknob	

Task 9.2

For each lever, describe where the fulcrum, the effort and the load are.

Lever	Fulcrum	Effort	Load
Seesaw	*the pivot in the middle*	*child pushing down on one end*	*child on the other end*
Wheelbarrow			
Pair of scissors			
Tweezers			

Task 9.3

Classify each lever as Class 1, 2 or 3. Use Figure 9.2 to help.

Lever	Class	Lever	Class
Seesaw	*1*	Wheelbarrow	*2*
Crowbar lifting a rock		Nutcracker	
Tweezers		Fishing rod	

Task 9.4
A lever has an effort arm of 1.5 m and a load arm of 0.3 m. Find its MA.

Task 9.5
A lever with an MA of 5 is used with an effort of 80 N. What load can it lift?

Task 9.6
A 600 N person sits 2 m from the pivot of a seesaw. How far from the pivot must a 400 N child sit to balance it?

Task 9.7
A crowbar has an MA of 8. What effort is needed to shift a 2400 N rock?

Task 9.8
A wheelbarrow carries a 400 N load positioned 0.4 m from the wheel. The handles are 1.2 m from the wheel.

(a) Identify the lever class.

(b) Calculate the mechanical advantage.

(c) Calculate the effort needed to lift the handles.

Task 9.9
Look at the trebuchet in Figure 9.3. Identify the effort, the load and the fulcrum, and explain why it is a Class 1 lever.

Task 9.10

A trebuchet has a long arm of 3.0 m and a short arm of 0.5 m.

(a) Calculate its mechanical advantage.

(b) The counterweight pushes with 500 N. What force acts on the projectile?

(c) How many times faster does the end of the long arm move than the counterweight?

Task 9.11

A door handle is always placed on the side of the door furthest from the hinges. Explain why, using the words *fulcrum*, *effort arm* and *mechanical advantage*.

Task 9.12

A student pushes down hard on a crowbar but cannot move a heavy rock. Suggest **two** changes to the way the crowbar is used that would help, and explain each one using mechanical advantage.

Task 9.13

Two ramps lead up to the same 1 m high stage. Ramp A is 2 m long and Ramp B is 4 m long.

(a) Which ramp needs less force to push a trolley up?

(b) On which ramp must the trolley be pushed further?

(c) Explain your answers using the idea of trading distance for force.

Part D • Above level

These questions go beyond the Year 7 standard. They ask you to combine ideas, design a lever, or judge someone else's reasoning.

Extension 9.14
A trebuchet has a short arm of 0.6 m and a long arm of 3.0 m.
(a) Calculate the mechanical advantage.
(b) The counterweight pushes with 600 N. Calculate the force at the projectile end.
(c) Explain why a trebuchet is designed with an MA less than 1, rather than greater than 1.

Extension 9.15
Two trebuchets have the same counterweight, dropping the same distance. Trebuchet A has arms of 0.5 m and 2.0 m; Trebuchet B has arms of 0.5 m and 3.5 m.
(a) Calculate the ratio long arm ÷ short arm for each.
(b) Explain which will launch the projectile faster, referring to the distance the arm tip travels.
(c) Suggest one practical limit on making the long arm even longer.

Extension 9.16
Your forearm is a Class 3 lever with the elbow as the fulcrum. Your biceps muscle pulls 4 cm from the elbow, and you hold a 20 N dumbbell 32 cm from the elbow.
(a) Calculate the mechanical advantage.
(b) Use the law of the lever to find the force the biceps must provide.
(c) Suggest why the body uses a lever with an MA less than 1 here.

Extension 9.17
On one side of a seesaw, a 300 N child sits 1.5 m from the pivot and a 200 N child sits 1.0 m from the pivot. Where must a 500 N adult sit on the other side to balance it? Show your working.

Extension 9.18
You need to lift a 1200 N log with a 2.5 m bar, but you can push with at most 300 N.
(a) What mechanical advantage do you need?
(b) Where should you place the fulcrum along the bar? Show your working.

Extension 9.19
A 600 N box must be loaded onto a truck 1 m high. Lifting it straight up uses $600 \times 1 = 600$ J of energy. Pushing it up a 3 m ramp takes a steady push of 250 N. The energy used is force $\times$ distance moved.
(a) Calculate the energy used pushing the box up the ramp.
(b) Calculate the efficiency of the ramp.
(c) Where did the extra energy go?

Extension 9.20
Ali says: *"A lever with an MA of 4 gives you four times as much energy as you put in."* Is Ali right? Explain, using the idea of trading distance for force.

10 Experiment 4: Mechanical Advantage

Prediction • Balancing a lever • Effort arm and load arm • Calculating MA

Victorian Curriculum 2.0: **VC2S8U13 • VC2S8I01 • VC2S8I02 • VC2S8I03 • VC2S8I04 • VC2S8I05**

Part A • Reading and comprehension

Reading — The Steelyard: Weighing with a Lever

Long before electronic scales, market sellers weighed goods with a *steelyard*. A steelyard is a lever: a long bar hanging from a hook near one end, which acts as the pivot. The goods hang from the short side, close to the pivot. A small, heavy weight slides along the long side, which is marked with a scale. The seller slides the weight out until the bar hangs level, then reads the weight of the goods off the scale.

How can a small weight balance a heavy load? The answer is distance. A lever balances when effort × effort arm equals load × load arm. Far from the pivot, a small weight turns the bar as strongly as a large weight close to it. A sliding weight ten times further from the pivot than the goods can balance goods ten times heavier than itself.

Steelyards are very old: bronze ones have been found at Pompeii, the Roman town buried by Vesuvius in AD 79. About three hundred years earlier, the Greek mathematician Archimedes had worked out the law of the lever that makes them work.

In this experiment you will do what a steelyard does: balance a heavy load near the pivot with a lighter mass further away, measure both arms, and work out your lever's mechanical advantage.

C1. Describe how a market seller uses a steelyard to find the weight of some goods.

C2. Explain how a small sliding weight can balance a much heavier load. Use the words *pivot* and *distance*.

C3. The sliding weight sits ten times further from the pivot than the goods. How heavy can the goods be, compared with the sliding weight?

C4. In this experiment, what will you balance, and what will you measure?

Part B • Planning the experiment

Experiment 4: Lever Mechanical Advantage

Curriculum: VC2S8U13 • VC2S8I01 • VC2S8I02 • VC2S8I03 • VC2S8I04 • VC2S8I05
Aim: *To investigate how the length of the effort arm affects the mechanical advantage of a lever.*

10.1 The Set-up

A metre stick rests on a fulcrum at its 50 cm mark, so the stick balances by itself. A 100 g **load** stands on the 30 cm mark, 20 cm from the fulcrum, and stays there for the whole experiment. On the other side you place an **effort** mass and slide it until the stick balances level. The distance from the fulcrum to the effort is the **effort arm**.

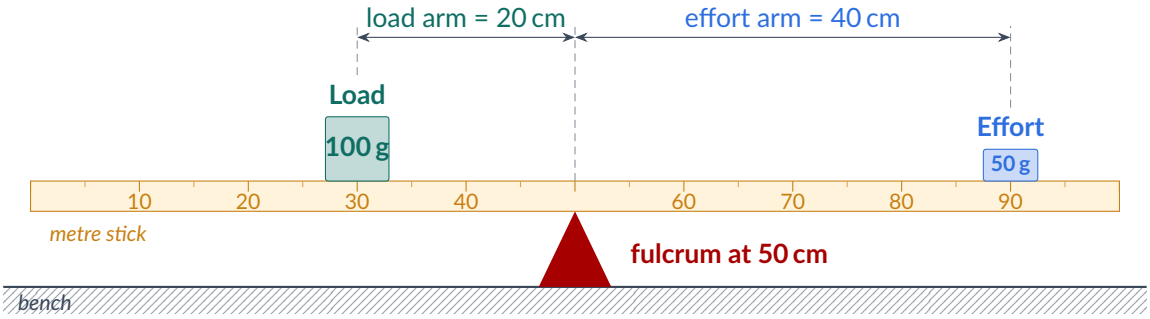


Figure E10.1 — The set-up. Measure each arm from the fulcrum to the centre of the mass. Here a 50 g effort balances the 100 g load at an effort arm of 40 cm.

10.2 Two Ways to Find the Mechanical Advantage

Mechanical advantage, two ways

From the arms: MA = effort arm ÷ load arm
From the forces: MA = load ÷ effort

Weight is proportional to mass ($W = mg$), so you can compare the forces using the masses in grams. When the lever is balanced, the two answers should agree.

Worked Example 10.1 — Mechanical advantage of a balanced lever

A 100 g load stands 20 cm from the fulcrum. A 50 g effort balances it 40 cm from the fulcrum. Find the MA from the arms and from the forces.

From the arms: $40 \div 20 = \mathbf{2}$
From the forces: $100 \div 50 = \mathbf{2}$

They agree. This is the law of the lever: $100 \times 20 = 50 \times 40 = 2000$ on both sides.

10.3 Making it a Fair Test

You will change the **effort mass** (50, 100, 150 and 200 g) and measure the **effort arm** needed to balance the stick. From these you calculate the MA. Everything else stays the same: the 100 g load, its place on the 30 cm mark, the fulcrum at the 50 cm mark, and the same metre stick.

Prediction — before you begin

As the effort mass increases from 50 g to 200 g, will the effort arm get longer or shorter? What will happen to the MA? Give a reason.

Equipment

Metre stick Triangular prism or thick pencil (fulcrum) One 100 g mass (load) Four 50 g slotted masses (effort) Sticky tack

Method

1. Rest the metre stick on the fulcrum at the 50 cm mark and check that it balances level. If it balances a little to one side of 50 cm, use that point as the fulcrum and measure every arm from there.
2. Stand the 100 g load on the 30 cm mark, so the load arm is 20 cm. Fix it in place with a little sticky tack. It stays there for the whole experiment.
3. Stand one 50 g mass on the other side of the fulcrum. Slide it until the stick balances level. Record the effort arm: the distance from the fulcrum to the centre of the mass.
4. Take the effort mass off and repeat step 3 twice more, so you have 3 trials.
5. Repeat steps 3–4 with an effort of 100 g, 150 g and 200 g (stack the 50 g masses).

Safety

Keep the masses low on the bench and hold the stick while you add or move a mass — an unbalanced stick can tip suddenly and drop the masses. Keep feet clear of the bench edge.

Notes

Part C • Results and analysis

Results

Load: 100 g *Load arm: 20 cm*

Effort mass (g)	Effort arm, trial 1 (cm)	Effort arm, trial 2 (cm)	Effort arm, trial 3 (cm)	Mean effort arm (cm)	MA = effort arm ÷ load arm	MA = load ÷ effort
e.g. 50	*39*	*41*	*40*	*40*	$40 \div 20 = 2$	$100 \div 50 = 2$
50						
100						
150						
200						

Calculating the mean and the MA

Mean: add the three trials and divide by 3. For 39, 41 and 40 cm: $(39 + 41 + 40) \div 3 = \mathbf{40}$ cm.
MA from the arms: $40 \div 20 = \mathbf{2}$. MA from the forces: $100 \div 50 = \mathbf{2}$.

Task 10.1

Identify the independent variable, the dependent variable, and two controlled variables in this experiment.

Task 10.2

What happened to the effort arm as the effort mass increased? What happened to the MA? Was your prediction correct?

Task 10.3

Compare your two MA columns. Do they agree? Suggest one reason why they might not agree exactly.

Task 10.4

Another group balanced the 100 g load (load arm 20 cm) with an 80 g effort. Use the law of the lever to predict their effort arm.

Task 10.5

A group's effort arms for 100 g were 19, 21 and 27 cm. Which value looks wrong? Find the mean of the other two.

Task 10.6

Use the law of the lever to explain *why* a heavier effort mass balanced the load closer to the fulcrum.

Task 10.7

Draw a line graph of your results. Put **effort mass (g)** on the horizontal axis and **mean effort arm (cm)** on the vertical axis. Plot each point, then draw a smooth curve of best fit — it will not be a straight line. Give your graph a title.

Part D • Above level

These questions go beyond the Year 7 standard: interpret your graph, use the law of the lever, or extend the method.

Extension 10.8
Your graph from Task 10.7 is a curve, not a straight line.
(a) For each row of your results, calculate effort mass × mean effort arm. What do you notice?
(b) Use the law of the lever to explain why this number should be the same every time, and what it should equal.

Extension 10.9
In this experiment the light effort sat on the long arm, so the MA was greater than 1. A trebuchet is a lever too.
(a) Describe how a trebuchet's effort and load are placed differently.
(b) Explain why a trebuchet is built with an MA *less* than 1.

Extension 10.10
The metre stick has its own weight, which acts at its centre — the 50 cm mark. Explain why the method keeps the fulcrum at the 50 cm mark, and predict what would go wrong if the fulcrum were moved to the 30 cm mark with nothing else changed.

Extension 10.11
Design a fair test to find out how the *load arm* affects the effort needed to balance the 100 g load. Use the same metre stick and masses.
(a) Name the independent and dependent variables.
(b) Name two variables you would keep the same.
(c) Predict what you would find, and give a reason using the law of the lever.

11 Levers Extension (Optional)

Year 8 extension • Turning effects • Arcs and tip speed • Energy and launch speed

Victorian Curriculum 2.0: **VC2S8U13 • VC2S8U15 • VC2S10U15 (above level)**

Optional lesson • Year 8 extension

Every part of this lesson goes beyond the Year 7 standard. Your teacher may skip it — Lesson 12 follows straight on from Lesson 10.

Part A • Reading and comprehension

Reading — Warwolf, the Biggest Trebuchet

In 1304 King Edward I of England laid siege to Stirling Castle in Scotland. For weeks his siege engines battered the walls. To finish the job, Edward ordered his chief engineer, Master James of St George, to build something far bigger: a trebuchet called *Warwolf*. It is believed to be the largest trebuchet ever made. Records say it took five master carpenters and forty-nine other labourers at least three months to build, and when it was taken apart it filled thirty wagons.

A trebuchet is a giant Class 1 lever. A heavy counterweight hangs from the short arm. When it is released it falls, and the long arm swings up and over, flinging a stone from a sling at its end. The counterweight only drops a few metres, but in the same moment the tip of the long arm sweeps through a huge arc — so the tip moves far faster than the counterweight.

The energy for the throw comes from the falling counterweight. Its gravitational potential energy becomes kinetic energy of the arm, the sling and the stone. The arms decide how that energy is shared out. A longer throwing arm moves its tip faster, but no arrangement of arms can give the stone more energy than the counterweight loses — and some is always wasted.

The castle's small garrison, only about thirty men, offered to surrender before Warwolf was ready. Edward refused to accept until he had seen his new machine work. One medieval chronicle claims that a single stone from Warwolf smashed through two of the castle's walls.

C1. Who ordered Warwolf to be built, and where and when was it used?

C2. Explain why a trebuchet is a Class 1 lever. Name its effort and its load.

C3. The counterweight only drops a few metres. Explain why the tip of the long arm moves so much faster than the counterweight.

C4. Where does the energy for the throw come from? Describe the energy transformation.

C5. Which phrase in the third paragraph tells you that a lever cannot create energy?

C6. Give two pieces of evidence from the passage that show how big Warwolf was.

C7. Suggest why Edward refused to accept the surrender until Warwolf had been tested.

C8. Suppose Warwolf's counterweight had been made heavier, with the arms kept the same. Using the passage, predict what would happen to the energy available and to the speed of the stone, and explain why.

Part B • Theory and examples

11.1 Turning Effects

A force that acts on a lever tries to turn it about the pivot. Its **turning effect** (also called its *moment*) depends on the size of the force and how far from the pivot it acts.

Turning effect

$$\text{turning effect} = \text{force} \times \text{distance from the pivot}$$

Force in newtons (N), distance in metres (m), turning effect in **newton metres (N m)**.

A lever balances when the total turning effect one way equals the total turning effect the other way. With two or more loads on one side, **add** their turning effects.

The law of the lever from Lesson 9 is a special case: effort × effort arm and load × load arm are the two turning effects.

Worked Example 11.1 — Two children on one side

On one side of a seesaw, a 300 N child sits 2.0 m from the pivot and a 200 N child sits 1.0 m from the pivot. Where must a 400 N child sit on the other side to balance it?

Left side: $300 \times 2.0 + 200 \times 1.0 = 600 + 200 = 800$ N m
Right side: $400 \times d = 800$ so $d = 800 \div 400 =$ **2.0** m

11.2 Arcs and Tip Speed

When a lever arm turns, every point on it turns through the **same angle in the same time**. A point far from the pivot travels around a bigger circle, so it covers a longer arc — and moves faster.

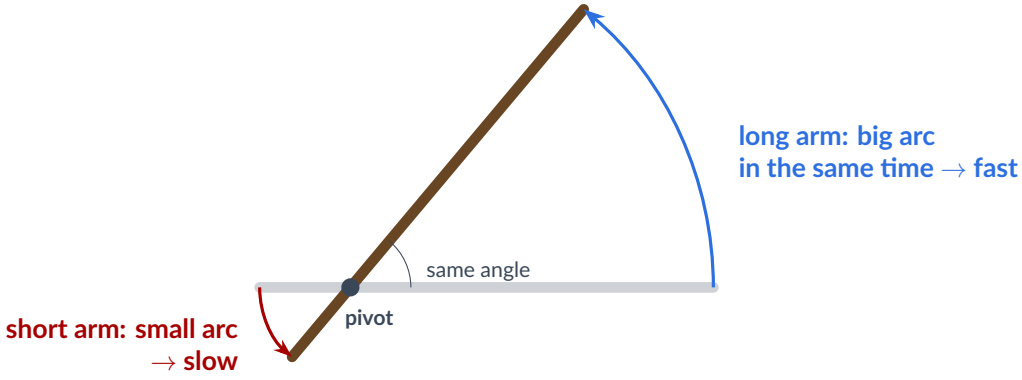


Figure 11.1 — Both ends of the arm turn through the same angle in the same time, but the end of the long arm travels a much longer arc.

Arc length and speed

$$\text{arc length} = \frac{\text{angle}}{360} \times 2 \times \pi \times r \qquad \text{average speed} = \frac{\text{arc length}}{\text{time}}$$

r = distance from the pivot (m) Use $\pi \approx 3.14$.

For two points on the same arm: $\dfrac{\text{tip speed}}{\text{counterweight speed}} = \dfrac{\text{long arm}}{\text{short arm}}$

Worked Example 11.2 — Speed of the arm tip

A trebuchet's long arm is 2.0 m long. During a launch it sweeps through 90° in 0.25 s. Find the arc travelled by the tip, and the tip's average speed.

Arc: $\frac{90}{360} \times 2 \times 3.14 \times 2.0 = 0.25 \times 12.56 = 3.14$ m
Speed: $3.14 \div 0.25 =$ **12.6** m/s

Worked Example 11.3 — Speed of the counterweight end

The same trebuchet has a short arm of 0.4 m. How fast does the counterweight end of the arm move?

Ratio: long arm ÷ short arm $= 2.0 \div 0.4 = 5$, so the tip moves 5 times as fast.
Speed: $12.6 \div 5 \approx$ **2.5** m/s

11.3 Energy Sets the Limit

The arms share out the speed, but the energy comes from the falling counterweight. The stone's kinetic energy can never be more than the gravitational potential energy the counterweight gives up — and because some energy is always wasted, it is always less.

From counterweight to launch speed

GPE lost $= m \times g \times h$ (counterweight mass and drop)
KE of stone = efficiency × GPE lost
KE $= \frac{1}{2}mv^2 \quad \Rightarrow \quad v = \sqrt{2 \times \text{KE} \div m}$ (stone mass)

Worked Example 11.4 — Launch speed from energy

A 20 kg counterweight falls 0.8 m. The trebuchet is 25% efficient and the stone has a mass of 0.2 kg. Find the stone's launch speed. Use $g = 9.8$ N/kg.

GPE lost: $20 \times 9.8 \times 0.8 = 156.8$ J
KE of stone: $0.25 \times 156.8 = 39.2$ J
Speed: $v^2 = 2 \times 39.2 \div 0.2 = 392$, so $v = \sqrt{392} \approx$ **19.8** m/s

Notes

Part C • Practice

Every question in this lesson goes beyond the Year 7 standard. Use $g = 9.8$ N/kg and $\pi \approx 3.14$. Show your working, and always include the unit.

Task 11.1
A 40 N force pushes on a door 0.5 m from its hinges. Find its turning effect.

Task 11.2
Which has the bigger turning effect: 60 N acting 0.4 m from a pivot, or 30 N acting 0.9 m from it? Show your working.

Task 11.3
A mechanic pushes on a spanner with 50 N, 0.25 m from the nut.

(a) Find the turning effect.

(b) What force would give the same turning effect if she pushed only 0.10 m from the nut?

Task 11.4
On one side of a seesaw, a 250 N child sits 1.6 m from the pivot and a 150 N child sits 1.0 m from the pivot. Where must a 500 N adult sit on the other side to balance it?

Task 11.5

Complete the table. Use arc length $= \frac{\text{angle}}{360} \times 2 \times \pi \times r$.

Angle turned	Distance from pivot r (m)	Arc length (m)
90°	2.0	$\frac{90}{360} \times 2 \times 3.14 \times 2.0 = 3.14$
90°	1.0	
180°	2.0	
60°	1.2	

Task 11.6

The tip of a trebuchet arm sweeps along a 4.8 m arc in 0.3 s. Find its average speed.

Task 11.7

A trebuchet has a long arm of 3.0 m and a short arm of 0.6 m. The counterweight end moves at 3 m/s. How fast does the tip move?

Task 11.8

Two trebuchets both have a short arm of 0.5 m. Trebuchet A has a long arm of 2.0 m; trebuchet B has a long arm of 3.0 m. In both, the counterweight end moves at 2.4 m/s.

(a) Find the tip speed of each trebuchet.

(b) Which one launches its stone faster, and why?

Task 11.9

A 15 kg counterweight falls 1.0 m. Find the GPE it gives up.

Task 11.10

A 0.1 kg stone leaves a trebuchet at 18 m/s. Find its kinetic energy.

Task 11.11

A counterweight gives up 150 J of GPE, and the stone gains 45 J of kinetic energy. Find the efficiency of the trebuchet, and name two places the rest of the energy went.

Task 11.12

A 12 kg counterweight falls 1.5 m. The trebuchet is 25% efficient, and the stone has a mass of 0.15 kg.

(a) Find the GPE the counterweight gives up.

(b) Find the KE given to the stone.

(c) Find the stone's launch speed.

Task 11.13

Explain, in your own words, why making the long arm longer makes the tip move faster, but cannot make the stone go faster and faster without any limit.

Part D • Challenge

The hardest questions in this extension lesson. They ask you to check a claim, combine turning effects with energy, or design a better trebuchet.

Extension 11.14
Sam says his model trebuchet, with a 2 kg counterweight that falls 0.5 m, launched a 0.05 kg ball at 25 m/s. Use energy to decide whether this is possible. Show your working.

Extension 11.15
A trebuchet has a short arm of 0.4 m. The builders want the tip to move 6 times as fast as the counterweight end.
(a) How long must the long arm be?
(b) Suggest one practical problem with making the long arm very long.

Extension 11.16
Before launch, a trebuchet arm is held level. A 30 kg counterweight hangs 0.5 m from the pivot on the short side. The arm's own weight, 100 N, acts 0.6 m from the pivot on the long side. A trigger rope pulls straight down on the end of the long arm, 2.5 m from the pivot. Find the force in the trigger rope.

Extension 11.17
During a launch, a trebuchet arm turns through 120° in 0.4 s. Its long arm is 2.4 m and its short arm is 0.6 m.
(a) Find the arc travelled by the tip, and its average speed.
(b) Find the average speed of the counterweight end.
(c) Check that your two speeds are in the same ratio as the arm lengths.

Extension 11.18
A 25 kg counterweight falls 1.2 m, and launches a 0.2 kg stone at 15 m/s.
(a) Calculate the efficiency of this trebuchet.
(b) Most of the energy did not reach the stone. Suggest where it went — think about what is still moving when the stone leaves the sling.

Extension 11.19
A counterweight gives up 200 J. The stone gets 60 J, the counterweight and arm are still moving with 90 J, and 50 J becomes thermal energy and sound. Draw a Sankey diagram using a scale of **1 cm = 20 J**. Label each arrow with its energy.

Draw your Sankey diagram here: energy in on the left, the useful energy straight on, the other energy turning down.

Extension 11.20
Real trebuchets fling the stone from a sling that hangs from the end of the long arm. Use the idea of arcs and tip speed to explain how the sling helps the stone leave faster.

12 Projectile Motion

Projectiles • Two independent motions • Time of flight • Measuring g

Victorian Curriculum 2.0: **VC2S8U14 • VC2S8H01 • VC2S10U17 (above level)**

Part A • Reading and comprehension

Reading — Galileo and the Two Motions

How does a cannonball fly? In 1537 the Italian mathematician Niccolò Tartaglia published one of the first books to use mathematics to study gunnery. He described a cannonball leaving the barrel in a straight line, then curving around in an arc, and finally dropping straight down to the ground. Gunnery books repeated his ideas for many years.

About a hundred years later, Galileo Galilei worked out what really happens, and he published it in 1638. He realised that a projectile is doing *two things at once*, and that the two motions do not affect each other. Sideways, once the ball is launched nothing pushes or pulls it (if we ignore the air), so it keeps moving sideways at a steady speed. Downwards, gravity pulls it, so it speeds up as it falls — exactly as if it had simply been dropped. Put a steady sideways motion together with a speeding-up downwards motion and you get a smooth curve called a *parabola*, with no straight sections at all.

The result that surprises people most is this. Launch one ball sideways and drop an identical ball from the same height at the same moment: they hit the ground at the same time. The launched ball lands far away, but its sideways speed makes no difference to how quickly gravity pulls it down.

That is why, in this unit, you can measure the strength of gravity just by timing how long a launched ball takes to fall. Galileo's idea replaced a model that had been trusted for a century — a good example of science changing when someone finds better evidence and better reasoning.

C1. Describe the path Tartaglia thought a cannonball followed.

C2. Galileo said a projectile does "two things at once". Describe each motion.

C3. According to the passage, why does a projectile's sideways speed stay the same?

C4. What is the name of the curved path that Galileo described?

C5. Explain why a ball launched sideways and a ball dropped from the same height land at the same time.

C6. Which phrase in the second paragraph tells you that the sideways motion and the downwards motion are independent?

C7. How does the passage show that scientific ideas can change? Give the old idea and the new one.

C8. Imagine Galileo's test on the Moon, where gravity is weaker and there is no air. Predict whether the launched and dropped balls would still land together, and whether they would take more or less time than on Earth. Explain.

Part B • Theory and examples

12.1 What is a Projectile?

A **projectile** is an object that has been launched and is then moving with only gravity acting on it. A thrown ball, water from a drinking fountain and a stone flung by a trebuchet are all projectiles. (We ignore air resistance, which is small for heavy, slow objects.) The curved path of a projectile is called its **trajectory**.

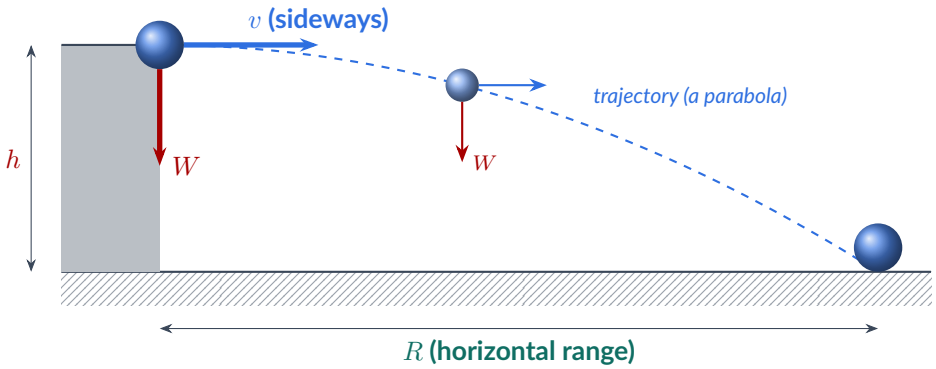


Figure 12.1 — A ball launched sideways from height h lands a horizontal distance R away. Once launched, the only force on it is its weight, W.

12.2 Two Independent Motions

	Sideways (horizontal)	Downwards (vertical)
Force acting	none (ignoring air)	gravity (the ball's weight)
Speed	stays the same	increases by 9.8 m/s every second
Distance	the same in every second	more in every second

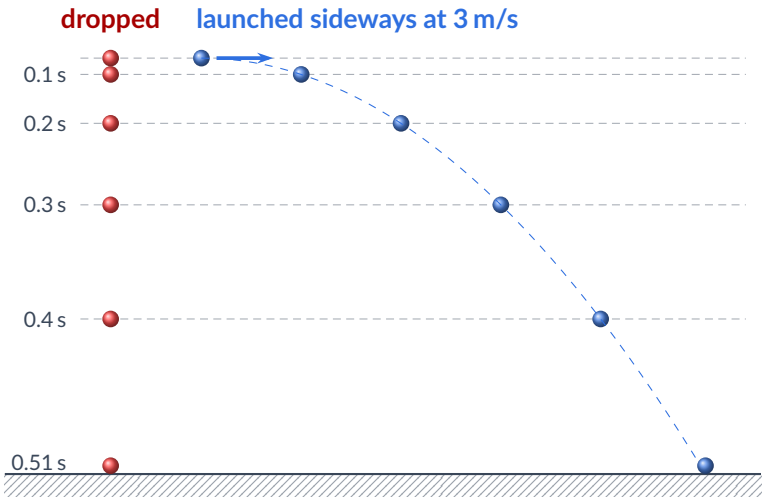


Figure 12.2 — Positions every 0.1 s. At every moment the launched ball is at the same height as the dropped ball, so they land together. Sideways, the launched ball moves the same distance in each 0.1 s.

12.3 Time of Flight

The time a projectile spends in the air depends only on the height it falls from, not on its sideways speed.

Key equations — launched sideways

$$h = \tfrac{1}{2}gt^2 \quad \Rightarrow \quad t = \sqrt{\frac{2h}{g}} \qquad R = v \times t \qquad g = \frac{2h}{t^2}$$

h = drop height (m) t = time of flight (s) $g = 9.8$ m/s^2
v = sideways launch speed (m/s) R = range (m)

Worked Example 12.1 — Time of flight

A ball rolls off a table 1.25 m high. How long does it take to land?

Step 1: $2h = 2 \times 1.25 = 2.5$
Step 2: $2.5 \div 9.8 = 0.255$
Step 3: $t = \sqrt{0.255} \approx$ **0.51** s

12.4 How Far Does It Go?

While the ball falls, it keeps moving sideways at its launch speed. So the distance it travels sideways is its launch speed multiplied by its time in the air.

Worked Example 12.2 — Range

The ball in Worked Example 12.1 leaves the table at 3.0 m/s. How far from the table does it land?

$R = v \times t = 3.0 \times 0.51 \approx$ **1.5** m

12.5 Measuring g

Rearranging $h = \tfrac{1}{2}gt^2$ gives $g = 2h \div t^2$. Measure the drop height and the time of flight, and you can calculate the strength of gravity. A slow-motion video gives the time:

$$t = \text{number of frames} \div \text{frames per second}$$

Worked Example 12.3 — Finding g from a video

A ball rolls off a bench 0.80 m high. A video filmed at 120 frames per second shows 48 frames between the ball leaving the bench and landing. Find t, then g.

Time: $t = 48 \div 120 = 0.40$ s
Gravity: $g = 2h \div t^2 = (2 \times 0.80) \div (0.40 \times 0.40) = 1.6 \div 0.16 =$ **10** m/s^2

This is close to the accepted value of 9.8 m/s^2.

Notes

Part C • Practice

Work through every task. Use $g = 9.8\,\text{m/s}^2$ and ignore air resistance. Show your working, and always include the unit.

Task 12.1

Is each object a projectile? Write **YES** or **NO** in the box.

Object	Projectile?	Object	Projectile?
A basketball after it leaves the player's hands	*YES*	A rocket with its engine on	*NO*
A stone flying from a trebuchet		A car driving along a road	
Water arcing from a drinking fountain		A ball rolling off the edge of a table	

Task 12.2

Complete the table for a ball that has been launched sideways.

	Sideways motion	Downwards motion
What force acts?	*none (ignoring air)*	
What happens to the speed?		

Task 12.3

One ball is dropped, and an identical ball is launched sideways from the same height at the same moment. The dropped ball lands after 0.6 s. When does the launched ball land? Explain.

Task 12.4

A ball rolls off a table 0.80 m high. Find its time of flight.

Task 12.5

A ball rolls off a ledge 2.0 m high. Find its time of flight.

Task 12.6
A ball is launched sideways at 4.0 m/s and is in the air for 0.50 s. How far does it travel sideways?

Task 12.7
A ball launched sideways lands 2.4 m away after 0.60 s. Find its launch speed.

Task 12.8
A ball falls 2.0 m in 0.64 s. Calculate g.

Task 12.9
A video filmed at 240 frames per second shows a fall lasting 60 frames. Find the time of the fall.

Task 12.10
A marble rolls off a desk 0.45 m high at 2.0 m/s.

(a) Find its time of flight.

(b) Find how far from the desk it lands.

Task 12.11
A group measures $g = 9.2$ m/s^2. Calculate their percentage error, using

$$\%\text{ error} = \frac{|\text{measured } g - 9.8|}{9.8} \times 100\%$$

Task 12.12

Two marbles roll off the same desk at the same moment, one at 1 m/s and one at 3 m/s.

(a) Which one lands first?

(b) Which one lands further from the desk?

(c) Explain both answers using the idea of two independent motions.

Task 12.13

Priya says: *"The faster I roll a ball off the table, the longer it stays in the air."* Is Priya right? Explain.

Part D • Above level

These questions go beyond the Year 7 standard. They ask you to combine equations, compare situations, or evaluate a method.

Extension 12.14
On the Moon, $g = 1.6\,\text{m/s}^2$. A ball rolls off a 1.25 m ledge at 2.0 m/s.
(a) Find its time of flight and how far it travels sideways.
(b) Compare your answers with the same roll on Earth ($t = 0.51$ s). Explain the difference.

Extension 12.15
Ball A rolls off a 0.45 m step and ball B rolls off a 1.80 m wall.
(a) Find the time of flight of each ball.
(b) B fell four times as far as A. How many times as long did it take? Use $h = \frac{1}{2}gt^2$ to explain why.

Extension 12.16
A ball rolls off a bench and takes 0.45 s to land. Use $h = \frac{1}{2}gt^2$ to find the height of the bench.

Extension 12.17
A ball machine fires tennis balls sideways from a height of 1.5 m at 20 m/s. Ignoring air resistance, how far do the balls travel before they land?

Extension 12.18
A group rolls a ball off a 0.80 m bench and measures a mean value of $g = 10.6\,\text{m/s}^2$.
(a) Calculate their percentage error.
(b) Suggest two sources of error in the method, and how each could be reduced.

Extension 12.19
In Worked Example 12.3, the fall lasted 48 frames at 120 frames per second.
(a) Suppose the student miscounted and recorded 49 frames. Calculate the new value of g.
(b) By how much did one frame change the answer? Explain why Experiment 5 and the trebuchet investigation film at 120 frames per second or more.

Extension 12.20
Galileo's idea says that sideways speed does not change how long a projectile takes to fall. Design a fair test, using a ramp, a table and a slow-motion video, to check this.
(a) Name the independent and dependent variables.
(b) Name two variables you would keep the same.
(c) Predict what you would find, and give a reason.